



US012397127B2

(12) **United States Patent**
Osborne et al.

(10) **Patent No.:** **US 12,397,127 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **RESPIRATORY HUMIDIFICATION SYSTEM**

(71) Applicant: **FISHER & PAYKEL HEALTHCARE LIMITED**, Auckland (NZ)

(72) Inventors: **Hamish Adrian Osborne**, Auckland (NZ); **Jonathan David Harwood**, Auckland (NZ); **Natalie May Robertson**, Auckland (NZ)

(73) Assignee: **FISHER & PAYKEL HEALTHCARE LIMITED**, Auckland (NZ)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 4 days.

(21) Appl. No.: **18/062,826**

(22) Filed: **Dec. 7, 2022**

(65) **Prior Publication Data**

US 2023/0166068 A1 Jun. 1, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/571,671, filed on Sep. 16, 2019, now Pat. No. 11,559,653, which is a
(Continued)

(51) **Int. Cl.**
A61M 16/08 (2006.01)
A61M 16/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61M 16/0808** (2013.01); **A61M 16/024** (2017.08); **A61M 16/0816** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC .. A61M 16/00; A61M 16/0057; A61M 16/08; A61M 16/0808; A61M 16/0816;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,085,833 A 2/1914 Wilson
1,154,259 A 9/1915 Light
(Continued)

FOREIGN PATENT DOCUMENTS

AU 667538 3/1996
AU 726022 10/2000
(Continued)

OTHER PUBLICATIONS

ISR for International App No. PCT/NZ2015/050011; dated Mar. 19, 2015, 4 pages.

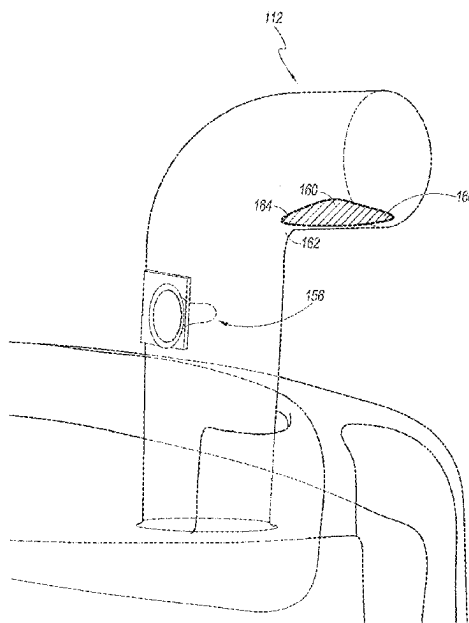
Primary Examiner — Joseph D. Boecker

(74) *Attorney, Agent, or Firm* — VIA LLP

(57) **ABSTRACT**

A system provides warm, humidified gas to a patient via a patient interface. Horizontal connections can be used between the humidification chamber and conduit. To reduce the likelihood of condensate flowing back to the humidification chamber, or dead space or gases recirculation regions occurring within the gases flow path, a raised portion is positioned inside of the flow path to improve flow characteristics and to provide a barrier for condensate back flow. The raised portion also reduces the amount of condensate that is formed in the system and provides better flow characteristics for sensing purposes.

19 Claims, 18 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/115,751, filed as application No. PCT/NZ2015/050011 on Feb. 9, 2015, now Pat. No. 10,449,319.

- (60) Provisional application No. 61/937,017, filed on Feb. 7, 2014.

(51) **Int. Cl.**

A61M 16/10 (2006.01)

A61M 16/16 (2006.01)

F15D 1/06 (2006.01)

F24F 13/08 (2006.01)

(52) **U.S. Cl.**

CPC *A61M 16/109* (2014.02); *A61M 16/16* (2013.01); *A61M 16/0875* (2013.01); *A61M 2205/3368* (2013.01); *F15D 1/06* (2013.01); *F24F 13/081* (2013.01)

(58) **Field of Classification Search**

CPC A61M 16/0875; A61M 16/109; A61M 16/16; A61M 2205/3368; F15D 1/04; F15D 1/06; F24F 13/081

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,073,335 A 3/1937 Connell
2,236,359 A 3/1941 Armstrong
2,510,125 A 6/1950 Meakin
2,516,864 A 8/1950 Gilmore et al.
2,590,797 A 3/1952 Siciliano
2,621,875 A 12/1952 Darling
2,634,311 A 4/1953 Darling
2,745,074 A 5/1956 Darling
3,117,596 A 1/1964 Kahn
3,163,707 A 12/1964 Darling
3,283,580 A 11/1966 Jacob et al.
3,394,954 A 7/1968 Sarns
3,404,684 A 10/1968 Brewer et al.
3,485,237 A 12/1969 Bedford
3,495,628 A 2/1970 Boender
3,582,094 A 6/1971 Whittaker
3,588,859 A 6/1971 Petree
3,623,511 A * 11/1971 Levin F15D 1/04
415/224
3,638,926 A 2/1972 Melville et al.
3,659,604 A 5/1972 Melville et al.
3,703,892 A 11/1972 Meyers
3,777,298 A 12/1973 Newman
3,806,102 A 4/1974 Valenta et al.
3,903,742 A 9/1975 Colton
3,945,378 A 3/1976 Paluch
3,954,920 A 5/1976 Heath
3,987,133 A 10/1976 Andra
3,990,727 A 11/1976 Gallagher
4,010,748 A 3/1977 Dobritz
4,028,444 A 6/1977 Brown
4,038,519 A 7/1977 Foucras
4,060,576 A 11/1977 Grant
4,098,853 A 7/1978 Brown et al.
4,111,197 A 9/1978 Warncke et al.
4,139,762 A 2/1979 Pohrer et al.
4,152,379 A 5/1979 Suhr
4,160,466 A 7/1979 Jousson
4,172,709 A 10/1979 Kippel et al.
4,183,248 A 1/1980 West
4,192,836 A 3/1980 Bartscher
4,301,200 A 11/1981 Langenfeld et al.
4,333,451 A 6/1982 Paluch
4,417,574 A 11/1983 Taloon et al.
4,428,403 A 1/1984 Lee et al.
4,463,593 A 8/1984 Parker

4,473,923 A 10/1984 Neroni et al.
4,529,867 A 7/1985 Velnosky et al.
4,531,551 A 7/1985 Eichelberger et al.
4,533,115 A 8/1985 Lissau
4,545,290 A 10/1985 Lieberman
4,558,708 A 12/1985 Labuda et al.
4,564,748 A 1/1986 Gupton
4,588,425 A 5/1986 Usry et al.
4,599,895 A 7/1986 Wiseman
4,621,632 A 11/1986 Bartels et al.
4,676,237 A 6/1987 Wood et al.
4,686,354 A 8/1987 Makin
4,708,831 A 11/1987 Elsworth et al.
4,714,078 A 12/1987 Paluch
4,753,758 A 6/1988 Miller
4,774,032 A 9/1988 Coates et al.
4,809,698 A 3/1989 Kogo
4,813,280 A 3/1989 Miller, Jr. et al.
4,830,515 A 5/1989 Cortes
4,844,512 A 7/1989 Gahwiler
4,942,877 A 7/1990 Sakai et al.
4,944,310 A 7/1990 Sullivan
4,967,744 A 11/1990 Chua
5,017,875 A 5/1991 Hori
5,027,811 A 7/1991 Tuxill
5,031,612 A 7/1991 Clementi
5,038,773 A 8/1991 Norlien
5,054,819 A 10/1991 Grunwald
5,058,588 A 10/1991 Kaestle
5,060,506 A 10/1991 Douglas
5,062,145 A 10/1991 Zwaan
5,109,471 A 4/1992 Lang
5,117,819 A 6/1992 Servidio et al.
5,125,069 A 6/1992 O'Boyle
5,127,442 A 7/1992 Blomqvist
5,134,996 A 8/1992 Bell
5,143,060 A 9/1992 Smith
5,148,801 A 9/1992 Douwens et al.
5,181,858 A 1/1993 Matz et al.
5,209,225 A 5/1993 Glenn
5,213,138 A 5/1993 Presz, Jr.
5,213,376 A 5/1993 Szabo
5,233,996 A 8/1993 Coleman et al.
5,279,796 A 1/1994 Parker
5,303,701 A 4/1994 Heins et al.
RE34,599 E 5/1994 Suszynski et al.
5,342,126 A 8/1994 Heston et al.
5,357,948 A 10/1994 Eilentropp
5,367,604 A 11/1994 Murray
5,392,770 A 2/1995 Clawson et al.
5,396,884 A 3/1995 Bagwell et al.
5,429,178 A 7/1995 Garey et al.
5,454,061 A 9/1995 Carlson
5,454,479 A 10/1995 Kraus
5,483,616 A 1/1996 Chiu et al.
5,484,223 A 1/1996 Saito
5,488,447 A 1/1996 Patton
5,495,872 A 3/1996 Gallagher et al.
5,499,737 A 3/1996 Kraus
RE35,225 E 4/1996 Herweck et al.
5,529,093 A 6/1996 Gallagher et al.
5,537,996 A 7/1996 McPhee
5,540,219 A 7/1996 Mechlenburg et al.
5,548,879 A 8/1996 Wu
5,551,883 A 9/1996 Davis
5,558,084 A 9/1996 Daniell et al.
5,564,415 A 10/1996 Dobson et al.
5,598,837 A 2/1997 Sirianne et al.
5,600,752 A 2/1997 Lopatinsky
5,623,940 A 4/1997 Daikuzono
5,630,806 A 5/1997 Inagaki et al.
5,637,006 A 6/1997 Almeras
5,640,951 A * 6/1997 Huddart F16L 11/12
128/911
5,660,567 A 8/1997 Nierlich et al.
5,667,306 A 9/1997 Montreuil et al.
5,720,293 A 2/1998 Nierlich et al.
5,778,872 A 7/1998 Fukunaga et al.
5,829,880 A 11/1998 Diedrich

(56)

References Cited

U.S. PATENT DOCUMENTS

5,881,393	A	3/1999	Marchello	7,137,654	B2	11/2006	Segal et al.
5,906,201	A	5/1999	Nilson	7,140,367	B2	11/2006	White et al.
5,913,249	A	6/1999	Weckstrom	7,157,035	B2	1/2007	Edirisuriya et al.
5,925,831	A	7/1999	Storsved	7,191,780	B2	3/2007	Faram
5,943,473	A	8/1999	Levine	7,225,809	B1	6/2007	Bowen et al.
5,975,591	A	11/1999	Guest	7,284,554	B2	10/2007	Shaw
5,979,247	A	11/1999	Kizawa	7,316,768	B2	1/2008	Aldridge et al.
D419,522	S	1/2000	Kamagai	7,316,769	B2	1/2008	Craighead
6,024,694	A	2/2000	Godlberg	7,327,547	B1	2/2008	Epstein
6,030,244	A	2/2000	Buckheit et al.	7,327,949	B1	2/2008	Cheng et al.
6,039,696	A	3/2000	Bell	7,334,587	B2	2/2008	Lake
6,050,260	A	4/2000	Daniell	7,364,436	B2	4/2008	Yen
6,053,482	A	4/2000	Glenn et al.	7,396,995	B2	7/2008	Laurent et al.
6,058,977	A	5/2000	Hotta	7,448,383	B2	11/2008	Delache et al.
6,062,244	A	5/2000	Arkans	7,469,586	B2	12/2008	Wild et al.
6,078,729	A	6/2000	Kopel	7,478,635	B2	1/2009	Wixey et al.
6,078,730	A	6/2000	Huddart et al.	7,525,663	B2	4/2009	Kwok et al.
6,095,505	A	8/2000	Miller	7,543,598	B1	6/2009	Hygema
6,102,037	A	8/2000	Koch	7,551,450	B2	6/2009	Sugawara et al.
6,105,970	A	8/2000	Siegrist et al.	7,607,360	B2	10/2009	Todokoro et al.
6,126,610	A	10/2000	Rich et al.	7,614,398	B2	11/2009	Virr et al.
6,128,963	A	10/2000	Bromster	7,637,288	B2	12/2009	Kressierer/Huber et al.
6,138,674	A	10/2000	Gull et al.	7,677,246	B2	3/2010	Kepler et al.
6,190,480	B1	2/2001	Carlson	7,743,767	B2	6/2010	Ging et al.
6,196,980	B1	3/2001	Akerfeldt et al.	7,766,050	B2	8/2010	Patel
6,201,983	B1	3/2001	Haumann et al.	7,794,426	B2	9/2010	Briones et al.
6,208,514	B1	3/2001	Stark	7,814,907	B2	10/2010	Bremner et al.
6,226,451	B1	5/2001	Wong	7,816,888	B2	10/2010	Rejman et al.
6,272,933	B1	8/2001	Gradon et al.	D628,288	S	11/2010	Row et al.
6,332,462	B1	12/2001	Krohn	7,827,981	B2	11/2010	Barnford
6,347,646	B2	2/2002	Fukui et al.	7,870,857	B2	1/2011	Dhuper et al.
6,349,722	B1	2/2002	Gradon et al.	7,874,291	B2	1/2011	Ging et al.
6,349,724	B1	2/2002	Burton et al.	7,900,528	B2	3/2011	Vincent
6,360,741	B2	3/2002	Truschel	7,913,689	B2	3/2011	Henry et al.
6,367,974	B1	4/2002	Lin	7,942,380	B2	5/2011	Bertinetti et al.
6,374,864	B1	4/2002	Philp	7,942,389	B2	5/2011	Koch et al.
6,394,145	B1	5/2002	Gessil	7,965,930	B2	6/2011	Carlson et al.
6,397,841	B1	6/2002	Kenyon et al.	7,983,542	B2	7/2011	McGhin et al.
6,402,207	B1	6/2002	Segal et al.	7,987,847	B2	8/2011	Wickham
6,435,180	B1	8/2002	Hewson et al.	7,992,554	B2	8/2011	Radomski et al.
6,467,477	B1	10/2002	Frank et al.	7,997,267	B2	8/2011	Ging et al.
6,508,249	B2	1/2003	Hoenig	8,025,849	B2	9/2011	Baldwin et al.
6,511,075	B1	1/2003	Schmidt	8,059,947	B2	11/2011	Bradley et al.
6,540,734	B1	4/2003	Chiu et al.	8,063,343	B2	11/2011	McGhin et al.
6,551,143	B2	4/2003	Tanaka et al.	8,078,040	B2	12/2011	Forrester
6,554,260	B1	4/2003	Lipscombe et al.	8,083,402	B2	12/2011	Mau et al.
6,584,972	B2	7/2003	McPhee	8,100,124	B2	1/2012	Becker et al.
6,591,061	B2	7/2003	Wang	8,122,882	B2	2/2012	McGhin et al.
6,598,604	B1	7/2003	Seakins	8,136,521	B2	3/2012	Matthews et al.
6,600,727	B1	7/2003	Mackay	8,137,082	B2	3/2012	Campbell
6,612,624	B1	9/2003	Segal et al.	8,181,940	B2	5/2012	Payne et al.
6,623,352	B2	9/2003	Illingworth	8,182,144	B2	5/2012	Koch
6,631,718	B1	10/2003	Lovell	8,186,345	B2	5/2012	Payton et al.
6,648,669	B1	11/2003	Kim et al.	8,186,352	B2	5/2012	Gunaratnam et al.
6,655,207	B1	12/2003	Speldrich et al.	8,197,123	B2	6/2012	Snyder et al.
6,655,975	B1	12/2003	Liedtke	8,206,337	B2	6/2012	Blackhurst et al.
6,668,828	B1	12/2003	Figley et al.	8,215,301	B2	7/2012	Richards et al.
6,685,491	B2	2/2004	Gergek	8,221,530	B2	7/2012	Peter et al.
6,698,966	B2	3/2004	Hilton et al.	8,226,293	B2	7/2012	Faries, Jr. et al.
6,824,180	B2	11/2004	Tomchak	8,240,306	B2	8/2012	Cortez, Jr. et al.
6,827,084	B2	12/2004	Grubb, Jr.	8,245,709	B2	8/2012	Rossen et al.
6,874,771	B2	4/2005	Birdsell et al.	8,245,710	B2	8/2012	Makinson et al.
6,895,803	B2	5/2005	Seakins et al.	8,253,076	B2	8/2012	Andel et al.
6,915,705	B1	7/2005	Truitt	8,257,286	B2	9/2012	Meyer et al.
6,918,389	B2	7/2005	Seakiins et al.	8,267,084	B2	9/2012	Kwok
6,935,337	B2	8/2005	Virr et al.	8,267,614	B2	9/2012	Khoe
6,943,566	B2	9/2005	Florin et al.	8,282,427	B2	10/2012	Yamazaki
6,953,354	B2	10/2005	Edirisuriya et al.	8,287,517	B2	10/2012	Hanlon et al.
7,017,818	B2	3/2006	Lebaschi et al.	8,316,848	B2	11/2012	Kwok et al.
7,043,979	B2	5/2006	Smith et al.	8,333,194	B2	12/2012	Hanlon et al.
7,063,668	B2	6/2006	Cardelius et al.	8,333,195	B2	12/2012	Cortez, Jr. et al.
7,086,422	B2	8/2006	Huber et al.	8,333,199	B2	12/2012	Landis et al.
7,090,541	B1	8/2006	Ho	8,355,753	B2	1/2013	Bochensko et al.
7,096,864	B1	8/2006	Mayer et al.	8,360,059	B2	1/2013	Koulechov et al.
7,120,354	B2	10/2006	Mackie et al.	8,365,726	B2	2/2013	Snow et al.
				8,381,724	B2	2/2013	Bowen et al.
				8,424,514	B2	4/2013	Oates et al.
				8,453,641	B2	6/2013	Payton et al.
				8,453,643	B2	6/2013	Sanchez et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,459,261 B2	6/2013	Ricciardelli	11,324,911 B2	5/2022	Osborne et al.
8,469,025 B2	6/2013	Mayer et al.	11,351,332 B2	6/2022	Mcintyre et al.
8,490,621 B2	7/2013	Radomski et al.	11,437,768 B2	9/2022	Scruggs et al.
8,496,001 B2	7/2013	Schermeier et al.	11,559,653 B2 *	1/2023	Osborne A61M 16/024
RE44,453 E	8/2013	Virr et al.	11,721,536 B2	8/2023	Osborne et al.
8,511,305 B2	8/2013	Liu et al.	11,801,360 B2	10/2023	Osborne et al.
8,511,651 B2	8/2013	Fridberg et al.	2001/0017134 A1	8/2001	Bahr
8,516,911 B2	8/2013	Inoue et al.	2001/0017880 A1	8/2001	Beerwerth et al.
8,522,782 B2	9/2013	Lewis et al.	2001/0050080 A1	12/2001	Seakins et al.
8,528,552 B2	9/2013	von Blumenthal	2002/0010032 A1	1/2002	Stiteler
8,544,465 B2	10/2013	Smith et al.	2002/0058436 A1	5/2002	Saba
8,545,096 B2	10/2013	Reiter	2002/0100320 A1 *	8/2002	Smith A61M 16/16 73/431
8,550,072 B2	10/2013	Thudor et al.	2002/0132511 A1	9/2002	Groebe et al.
8,631,789 B2	1/2014	Virr et al.	2002/0153011 A1	10/2002	Tanhehco
8,640,560 B2	2/2014	Burke	2003/0066526 A1	4/2003	Thurdor et al.
8,640,696 B2	2/2014	Pujol et al.	2003/0066530 A1	4/2003	Shahbazzpour et al.
8,651,800 B2	2/2014	Li	2003/0107325 A1	6/2003	Birkhead
8,733,348 B2	5/2014	Korneff et al.	2003/0127096 A1	7/2003	McAuliffe
8,733,349 B2	5/2014	Bath et al.	2003/0148664 A1	8/2003	Cheng
8,756,990 B2	6/2014	Speldrich	2003/0154977 A1	8/2003	White et al.
8,770,190 B2	7/2014	Doherty et al.	2003/0183294 A1	10/2003	Carlson
8,776,790 B2	7/2014	Gentner et al.	2003/0200727 A1	10/2003	Kim
8,783,252 B2	7/2014	Pierro et al.	2003/0236015 A1	12/2003	Edirisuriya et al.
8,800,970 B2	8/2014	Heine et al.	2004/0013162 A1	1/2004	Beerwerth et al.
8,844,388 B2	9/2014	Burke	2004/0055597 A1	3/2004	Virr et al.
8,844,521 B2	9/2014	McCarthy	2004/0060558 A1	4/2004	Gradon et al.
8,851,071 B2	10/2014	Kuo et al.	2004/0074493 A1	4/2004	Seakins et al.
8,905,023 B2	12/2014	Niland et al.	2004/0079371 A1	4/2004	Gray
8,915,250 B2	12/2014	Dugan et al.	2004/0087213 A1	5/2004	Kao
8,931,481 B2	1/2015	Jones et al.	2004/0099268 A1	5/2004	Smith et al.
8,939,147 B2	1/2015	Henry et al.	2004/0101026 A1	5/2004	Nitta et al.
8,960,727 B2	2/2015	Kendrick	2004/0149284 A1	8/2004	Smith et al.
8,985,105 B2	3/2015	Burton et al.	2004/0168530 A1	9/2004	Adolfs
8,997,740 B2	4/2015	Pujol et al.	2004/0182392 A1	9/2004	Gerder et al.
9,022,946 B2	5/2015	Haque	2004/0221843 A1	11/2004	Baecke
9,039,277 B2	5/2015	Le Bouquin et al.	2004/0234254 A1	11/2004	Czupich et al.
9,067,036 B2	6/2015	Kornell et al.	2004/0239001 A1	12/2004	Edirisuriya et al.
9,095,668 B2	8/2015	Blackhurst et al.	2004/0244858 A1	12/2004	Jeong
9,119,933 B2	9/2015	Bedford et al.	2005/0039809 A1	2/2005	Speldrich
9,132,252 B2	9/2015	Barlow et al.	2005/0059957 A1	3/2005	Campbell et al.
9,162,035 B2	10/2015	Kwok	2005/0152733 A1	7/2005	Patel
9,186,477 B2	11/2015	Hunt et al.	2006/0030191 A1	2/2006	Tuin et al.
9,205,220 B2	12/2015	Korneff et al.	2006/0118113 A1	6/2006	Bremner et al.
9,212,673 B2	12/2015	Korneff et al.	2006/0137445 A1	6/2006	Smith et al.
9,242,064 B2	1/2016	Rustad et al.	2006/0150712 A1	7/2006	Berstis et al.
9,254,368 B2	2/2016	Blumental et al.	2006/0165829 A1	7/2006	Smith et al.
9,289,572 B2	3/2016	Korneff et al.	2006/0196510 A1	9/2006	McDonald et al.
RE46,079 E	7/2016	Virr et al.	2006/0237012 A1	10/2006	Thurdor et al.
9,381,317 B2	7/2016	Landis et al.	2006/0249160 A1	11/2006	Scarberry
9,387,299 B2	7/2016	Zwolinsky et al.	2006/0283450 A1	12/2006	Shissler et al.
9,427,547 B2	8/2016	Landis et al.	2007/0039374 A1	2/2007	Borali
9,446,210 B2	9/2016	Orr et al.	2007/0079982 A1	4/2007	Laurent et al.
9,512,856 B2	12/2016	Nibu et al.	2007/0107737 A1	5/2007	Landis et al.
9,517,321 B2	12/2016	Buechi et al.	2007/0110124 A1	5/2007	Shiraki et al.
9,545,493 B2	1/2017	Mayer et al.	2007/0144519 A1	6/2007	Henry et al.
9,566,409 B2	2/2017	Gründler et al.	2007/0157928 A1	7/2007	Pujol et al.
9,572,949 B2	2/2017	Vos et al.	2007/0169776 A1	7/2007	Kepler et al.
9,572,951 B2	2/2017	Barker et al.	2007/0175473 A1	8/2007	Lewis et al.
9,586,019 B2	3/2017	Heine et al.	2007/0193580 A1	8/2007	Feldhahn et al.
9,642,979 B2	5/2017	Korneff et al.	2007/0248934 A1	10/2007	Mosimann
RE46,571 E	10/2017	Virr et al.	2007/0272240 A1	11/2007	Aylsworth et al.
9,838,759 B2	12/2017	Kirmse et al.	2007/0284361 A1	12/2007	Nadjafizadeh et al.
9,861,778 B2	1/2018	Roderick et al.	2008/0000474 A1	1/2008	Jochle et al.
9,937,314 B2	4/2018	Buechi et al.	2008/0015257 A1	1/2008	Grosskreutz et al.
9,937,316 B2	4/2018	Buechi et al.	2008/0027344 A1	1/2008	Terry
9,974,921 B2	5/2018	Klenner et al.	2008/0028850 A1	2/2008	Payton et al.
9,987,455 B2	6/2018	Stoks et al.	2008/0051674 A1	2/2008	Davenport et al.
10,046,136 B2	8/2018	Pujol	2008/0053456 A1	3/2008	Brown
10,105,511 B2	10/2018	Buechi	2008/0066741 A1	3/2008	LeMahieu et al.
10,245,407 B2	4/2019	Osborne et al.	2008/0066751 A1	3/2008	Polacsek
10,449,319 B2 *	10/2019	Osborne A61M 16/024	2008/0072900 A1	3/2008	Kenyon et al.
10,772,547 B1	9/2020	Lee et al.	2008/0077063 A1	3/2008	Meyer et al.
10,828,482 B2	11/2020	Osborne et al.	2008/0105257 A1	5/2008	Klasek et al.
10,974,015 B2	4/2021	Stoks et al.	2008/0142019 A1	6/2008	Lewis et al.
11,129,956 B2	9/2021	Klenner et al.	2008/0196716 A1	8/2008	Wachter
			2008/0202512 A1	8/2008	Kressierer
			2008/0205481 A1	8/2008	Faries
			2008/0205979 A1	8/2008	Gombert et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0207028	A1	8/2008	Schutz	2013/0008158	A1	1/2013	Hon
2008/0251073	A1	10/2008	Jassell et al.	2013/0042867	A1	2/2013	Kwok et al.
2008/0253427	A1	10/2008	Kamen et al.	2013/0043677	A1	2/2013	Gibson
2008/0264413	A1	10/2008	Doherty et al.	2013/0079667	A1	3/2013	Berkan et al.
2008/0295847	A1	12/2008	Gobel	2013/0081619	A1	4/2013	Seakins et al.
2008/0302362	A1	12/2008	Kwok	2013/0087143	A1	4/2013	Pujol
2008/0308169	A1	12/2008	Nielsen et al.	2013/0104888	A1	5/2013	Landis et al.
2009/0041080	A1	2/2009	Koch	2013/0104901	A1	5/2013	Landis et al.
2009/0044808	A1	2/2009	Guney et al.	2013/0112201	A1	5/2013	Graham et al.
2009/0050150	A1	2/2009	Rossen et al.	2013/0112202	A1	5/2013	Fogelbrink
2009/0056712	A1	3/2009	Cortez, Jr. et al.	2013/0152931	A1	6/2013	Sims et al.
2009/0056713	A1	3/2009	Cortez, Jr. et al.	2013/0174839	A1	7/2013	Ging et al.
2009/0078259	A1	3/2009	Kooji et al.	2013/0199529	A1	8/2013	Power et al.
2009/0087259	A1	4/2009	Bettinger	2013/0206140	A1	8/2013	Kepler et al.
2009/0107493	A1	4/2009	Liu et al.	2013/0208765	A1	8/2013	Takahashi
2009/0107496	A1	4/2009	McGhin et al.	2013/0237781	A1	9/2013	Gym
2009/0107501	A1	4/2009	Krieger	2013/0239960	A1	9/2013	Bertinetti et al.
2009/0107981	A1	4/2009	Andel et al.	2013/0239966	A1	9/2013	Klasek et al.
2009/0110022	A1	4/2009	Snyder et al.	2013/0247905	A1	9/2013	Miller et al.
2009/0110378	A1	4/2009	Bradley et al.	2013/0248044	A1	9/2013	Shiga et al.
2009/0135884	A1	5/2009	Sisk	2013/0252461	A1	9/2013	Gross
2009/0174092	A1	7/2009	Kwok et al.	2013/0255677	A1	10/2013	Varga
2009/0180829	A1	7/2009	Rejman et al.	2013/0333701	A1	12/2013	Herron
2009/0223514	A1	9/2009	Smith et al.	2013/0340752	A1	12/2013	Landis et al.
2009/0243226	A1	10/2009	Liebold	2014/0001658	A1	1/2014	Virr
2009/0247989	A1	10/2009	Burke	2014/0007872	A1	1/2014	Grundler et al.
2009/0301482	A1	12/2009	Burton et al.	2014/0020684	A1	1/2014	Klasek et al.
2009/0320840	A1	12/2009	Klasek et al.	2014/0048065	A1	2/2014	Haroutunian
2010/0015830	A1	1/2010	Simeon et al.	2014/0090649	A1	4/2014	Groll et al.
2010/0083965	A1	4/2010	Virr et al.	2014/0116433	A1	5/2014	Ghalib et al.
2010/0087749	A1	4/2010	Tovey	2014/0130802	A1	5/2014	Virr et al.
2010/0102799	A1	4/2010	Schmidrig	2014/0202460	A1	7/2014	Bath et al.
2010/0116272	A1	5/2010	Row et al.	2014/0202462	A1	7/2014	Stoks et al.
2010/0147301	A1	6/2010	Kwok	2014/0202463	A1	7/2014	Ging et al.
2010/0154796	A1	6/2010	Smith et al.	2014/0216446	A1	8/2014	Wruck
2010/0204602	A1	8/2010	Addington et al.	2014/0246021	A1	9/2014	Buechi et al.
2010/0242963	A1	9/2010	Brieger et al.	2014/0251322	A1	9/2014	Miller
2010/0272507	A1	10/2010	Khoe	2014/0251331	A1	9/2014	Korneff et al.
2010/0282247	A1	11/2010	Kadrichu et al.	2014/0283829	A1	9/2014	Miller
2010/0292601	A1	11/2010	Dompeling et al.	2014/0311489	A1	10/2014	Heine et al.
2011/0017212	A1	1/2011	Kenyon et al.	2014/0318536	A1	10/2014	Landis et al.
2011/0023874	A1	2/2011	Bath et al.	2014/0331786	A1	11/2014	Romano
2011/0046433	A1	2/2011	Khodak	2014/0338666	A1	11/2014	Visveshwara et al.
2011/0046494	A1	2/2011	Balji et al.	2014/0345614	A1	11/2014	Kwok
2011/0073109	A1	3/2011	Mayer et al.	2014/0366876	A1	12/2014	Huby et al.
2011/0078109	A1	3/2011	Griggs	2015/0027204	A1	1/2015	Stoks et al.
2011/0088693	A1	4/2011	Somervell et al.	2015/0040897	A1	2/2015	Buechi
2011/0108028	A1	5/2011	Zollinger	2015/0048530	A1	2/2015	Cheung et al.
2011/0108031	A1	5/2011	Korneff et al.	2015/0059745	A1	3/2015	Barker et al.
2011/0114093	A1	5/2011	Patil et al.	2015/0083126	A1	3/2015	Rogers
2011/0155132	A1	6/2011	Virr et al.	2015/0083132	A1	3/2015	Jones et al.
2011/0156289	A1	6/2011	Steg et al.	2015/0090260	A1	4/2015	Seakins et al.
2011/0180068	A1	7/2011	Kenyon et al.	2015/0096560	A1	4/2015	Klenner et al.
2011/0184264	A1	7/2011	Galasso et al.	2015/0107588	A1	4/2015	Cheung et al.
2011/0186048	A1	8/2011	Casse et al.	2015/0144130	A1	5/2015	O'Donnell et al.
2011/0247623	A1	10/2011	McCarthy	2015/0185093	A1	7/2015	Kitzman
2011/0253136	A1	10/2011	Sweeney et al.	2015/0196725	A1	7/2015	Oates et al.
2011/0283999	A2	11/2011	Smith et al.	2015/0359990	A1	12/2015	Barker et al.
2011/0308518	A1	12/2011	McGroary et al.	2016/0008560	A1	1/2016	Kwok
2011/0313689	A1	12/2011	Holley et al.	2016/0015937	A1	1/2016	Winski et al.
2011/0316562	A1	12/2011	Cefai et al.	2016/0022954	A1	1/2016	Bath et al.
2012/0012108	A1	1/2012	Sata et al.	2016/0051789	A1	2/2016	Korneff et al.
2012/0060838	A1	3/2012	Lapoint et al.	2016/0089510	A1	3/2016	Korneff et al.
2012/0073573	A1	3/2012	Thurdor et al.	2016/0101258	A1	4/2016	Rustad et al.
2012/0125333	A1	5/2012	Bedford et al.	2016/0199612	A1	7/2016	Foot et al.
2012/0125334	A1	5/2012	Korneff et al.	2016/0256642	A1	9/2016	Soysa et al.
2012/0146251	A1	6/2012	Heine et al.	2016/0256657	A1	9/2016	Klasek et al.
2012/0174924	A1	7/2012	Smith et al.	2016/0296721	A1	10/2016	Landis et al.
2012/0215125	A1	8/2012	Orr et al.	2016/0310691	A1	10/2016	Bath et al.
2012/0227738	A1	9/2012	Virr et al.	2016/0367776	A1	12/2016	Landis et al.
2012/0234323	A1	9/2012	Connor	2016/0367779	A1	12/2016	Landis et al.
2012/0255758	A1	10/2012	Lee	2017/0000968	A1	1/2017	Harrington et al.
2012/0266880	A1	10/2012	Young	2017/0095635	A1	4/2017	Huby et al.
2012/0283570	A1	11/2012	Tegg	2017/0136198	A1	5/2017	Delangre et al.
2012/0285448	A1	11/2012	Dugan et al.	2017/0151411	A1	6/2017	Osborne et al.
				2017/0161461	A1	6/2017	Delangre et al.
				2017/0173293	A1	6/2017	Osborne et al.
				2017/0197057	A1	7/2017	Osborne et al.
				2017/0239432	A1	8/2017	Delangre et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0326320	A1	11/2017	Baigent et al.
2018/0078730	A1	3/2018	Bath et al.
2018/0169361	A1	6/2018	Dennis et al.
2018/0214660	A1	8/2018	Stoks et al.
2018/0250491	A1	9/2018	Row et al.
2018/0296791	A1	10/2018	Klenner et al.
2019/0255278	A1	8/2019	Osborne et al.
2020/0061329	A1	2/2020	Mcintyre et al.
2020/0101253	A1	4/2020	Osborne et al.
2021/0220601	A1	7/2021	Stoks et al.
2022/0031993	A1	2/2022	Klenner et al.
2022/0280744	A1	9/2022	Osborne et al.
2022/0313940	A1	10/2022	Mcintyre et al.

FOREIGN PATENT DOCUMENTS

AU	7179100	3/2001
AU	2000071791	3/2001
AU	2001/028104	9/2001
AU	2007317198	5/2008
AU	2010206053	2/2011
AU	2013201490	4/2013
CA	1202862	4/1986
CA	2464530	5/2003
CA	2495451	3/2004
CA	2535974	10/2011
CA	2393743	1/2012
CA	2852215	4/2013
CN	2243015	12/1996
CN	1598510	3/2005
CN	1688358	10/2005
CN	101541367	9/2009
CN	101666664	3/2010
CN	102844645	12/2012
CN	201672170	12/2015
CN	106535971	12/2020
DE	3110903	9/1982
DE	3618614	12/1987
DE	4020522	1/1992
DE	4102223	7/1992
DE	19647548	5/1998
DE	19958296	9/2001
DE	20 2004 006 484.7	9/2005
DE	1020040307 47	1/2006
DE	20 2005 008 152.3	10/2006
DE	20 2005 008 156.6	10/2006
DE	203 21 468.4	8/2007
DE	203 21 469.2	8/2007
DE	203 21 470.6	8/2007
DE	203 21 471.4	8/2007
DE	203 21 472.2	8/2007
DE	20 2006 007 397.3	9/2007
DE	20 2004 021 759.7	10/2007
DE	20 2006 011 754.7	12/2007
DE	201 22 844.0	5/2008
DE	102007003454	7/2008
DE	102007003455	8/2008
DE	102007039391	2/2009
DE	102008001022	10/2009
DE	20 2004 021 757.0	9/2010
DE	20 2004 021 758.9	9/2010
DE	201 22 937.4	9/2010
DE	20 2004 021 756.2	10/2010
DE	20 2004 021 774.0	11/2010
DE	20 2004 021 777.5	12/2010
DE	20 2004 021 794.5	2/2011
DE	20 2004 021 795.3	2/2011
DE	20 2004 021 796.1	2/2011
DE	20 2004 021 798.8	2/2011
DE	20 2006 020 951.4	2/2011
DE	20 2006 020 952.4	2/2011
DE	20 2004 021829.1	5/2011
DE	201 22 943.9	5/2011
DE	201 22 944.7	5/2011
DE	201 22 945.5	5/2011

DE	20 2005 021 927.4	6/2011
DE	20 2006 021 019.9	11/2011
DE	203 21 882.5	12/2011
DE	20 2004 021876.3	1/2012
DE	20 2007 019350.5	1/2012
DE	20 2011 107 902.7	1/2012
DE	20 2010 016 037.5	3/2012
DE	20 2012 007 229.3	10/2012
EP	0111248	6/1984
EP	0050984	12/1984
EP	0201985	11/1986
EP	0291921	11/1988
EP	0535952	4/1993
EP	0567158	10/1993
EP	0232864	5/1994
EP	0672430	9/1995
EP	0885623	12/1998
EP	1262208	12/2002
EP	1352670	10/2003
EP	1396277	3/2004
EP	1535722	6/2005
EP	1606000	12/2005
EP	1634614	3/2006
EP	1646910	4/2006
EP	1129743	5/2006
EP	1669098	6/2006
EP	1035887	7/2006
EP	1683066	7/2006
EP	1457223	10/2006
EP	1741462	1/2007
EP	1837640	9/2007
EP	1055431	11/2007
EP	1924311	5/2008
EP	1933914	6/2008
EP	1979030	10/2008
EP	2079505	7/2009
EP	2089086	8/2009
EP	2101851	9/2009
EP	2236167	10/2010
EP	2307082	4/2011
EP	2335761	6/2011
EP	2340867	7/2011
EP	2355881	8/2011
EP	2133611	9/2011
EP	2415445	2/2012
EP	2471568	7/2012
EP	2281138	10/2012
EP	2514478	10/2012
EP	2575944	4/2013
EP	2613840	7/2013
EP	2629080	8/2013
EP	2640451	9/2013
EP	2651481	10/2013
EP	2522255	3/2014
EP	2703034	3/2014
EP	2760516	8/2014
EP	2830695	2/2015
EP	2877224	6/2015
EP	3013402	5/2016
EP	1359962	8/2016
EP	3053623	8/2016
EP	3148418	4/2017
EP	3082920	10/2017
EP	3148419	1/2018
GB	1310949	3/1973
GB	1364127	8/1974
GB	2176313	12/1986
GB	2224957	5/1990
GB	2504284	1/2014
IN	2005000354	5/2006
JP	59113392	6/1984
JP	S63-161973	7/1988
JP	H0623051	2/1994
JP	H11-000398	1/1999
JP	11033119	2/1999
JP	11286058	10/1999
JP	2000-005314	1/2000
JP	2001095920	4/2001
JP	2001-129091	5/2001

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	20011511507	8/2001
JP	2003139276	5/2003
JP	03194747	7/2003
JP	2003275312	9/2003
JP	2004-507889	3/2004
JP	2005/331101 A	12/2005
JP	2006-504469	2/2006
JP	2006-506128	2/2006
JP	2008132370	6/2008
JP	4242816	3/2009
JP	2009-523035	6/2009
JP	2010-027246	2/2010
JP	44022293	2/2010
JP	2010-194130	9/2010
JP	2010-256993	11/2010
JP	2011511507	4/2011
JP	2011-118836	6/2011
JP	2011125618	6/2011
JP	11248076	12/2011
JP	2012-134934	7/2012
JP	2013-524989	6/2013
JP	H 05208935	6/2013
NZ	564886	2/2011
NZ	587113	12/2011
NZ	586325	1/2012
NZ	597020	6/2013
NZ	600986	8/2013
NZ	604137	6/2014
NZ	610299	11/2014
NZ	630762	2/2016
NZ	625605	4/2016
NZ	710078	1/2017
NZ	710351	1/2017
NZ	733931	2/2019
TW	201245604	11/2012
WO	WO 1996/020748	7/1996
WO	WO 97/18001	5/1997
WO	WO 1997/042475	11/1997
WO	WO 2000/029057	5/2000
WO	WO 2001/032069	5/2001
WO	WO 01/41854	6/2001
WO	WO 01/97894	12/2001
WO	WO 2002/017991	3/2002
WO	WO 02/066106	8/2002
WO	WO 02/066107	8/2002
WO	WO 2002/066106	8/2002
WO	WO 2002/075854	9/2002
WO	WO 2003/026721	4/2003
WO	WO 2004/011072	2/2004
WO	WO 2004/024429	3/2004
WO	WO 2004/037330	5/2004
WO	2004093955 A	11/2004
WO	WO 2004/092955	11/2004
WO	WO 2004/093954	11/2004
WO	WO 2005/011785	2/2005
WO	WO 2005/021076	3/2005
WO	WO 2005/079670	9/2005

WO	WO 2006/017350	2/2006
WO	WO 2006/019323	2/2006
WO	WO 2007/019626	2/2007
WO	WO 2007/043060	4/2007
WO	WO 2007/051230	5/2007
WO	WO 2008/011220	1/2008
WO	WO 2008/055308	5/2008
WO	WO 2008/056993	5/2008
WO	WO 2008/058328	5/2008
WO	WO 2008/060046	5/2008
WO	WO 2008/060295	5/2008
WO	WO 2008/076230	6/2008
WO	WO 2008/097211	12/2008
WO	WO 2009/002004	12/2008
WO	WO 2009/006586	1/2009
WO	WO 2009/022004	2/2009
WO	WO 2009/127192	10/2009
WO	WO 2009/146484	12/2009
WO	WO 2010/031125	3/2010
WO	WO 2010/031126	3/2010
WO	WO 2010/091259	8/2010
WO	WO 2011/021708	2/2011
WO	WO 2011/030251	3/2011
WO	WO 2011/059622	5/2011
WO	WO 2011/059623	5/2011
WO	WO 2012/053910	4/2012
WO	WO 2012/065999	5/2012
WO	WO 2012/087644	6/2012
WO	WO 2012/100291	8/2012
WO	WO 2012/154883	11/2012
WO	WO 2012/164407	12/2012
WO	WO 2013/022356	2/2013
WO	WO 2013/026901	2/2013
WO	WO 2013/045572	4/2013
WO	WO 2013/045575	4/2013
WO	WO 2013/045586	4/2013
WO	WO 2013/049660	4/2013
WO	WO 2013/050907	4/2013
WO	WO 2013/088351	6/2013
WO	WO 2013/127474	9/2013
WO	WO 2013/137753	9/2013
WO	WO-2013137753 A1 *	9/2013
WO	WO 2013/151447	10/2013
WO	WO 2013/162386	10/2013
WO	WO 2014/015382	1/2014
WO	WO 2014/055407	4/2014
WO	WO 2014/077706	5/2014
WO	WO 2014/138804	9/2014
WO	WO 2014/205513	12/2014
WO	WO 2015/038013	3/2015
WO	WO 2015/060729	4/2015
WO	WO 2015/119515	8/2015
WO	WO 2015/160268	10/2015
WO	WO 2015/179916	12/2015
WO	WO 2016/042522	3/2016
WO	WO 2016/089224	6/2016
WO	WO 2016/139645	6/2016
WO	WO 2017/027906	2/2017
WO	WO 2017/126980	7/2017

* cited by examiner

..... A61M 16/0003

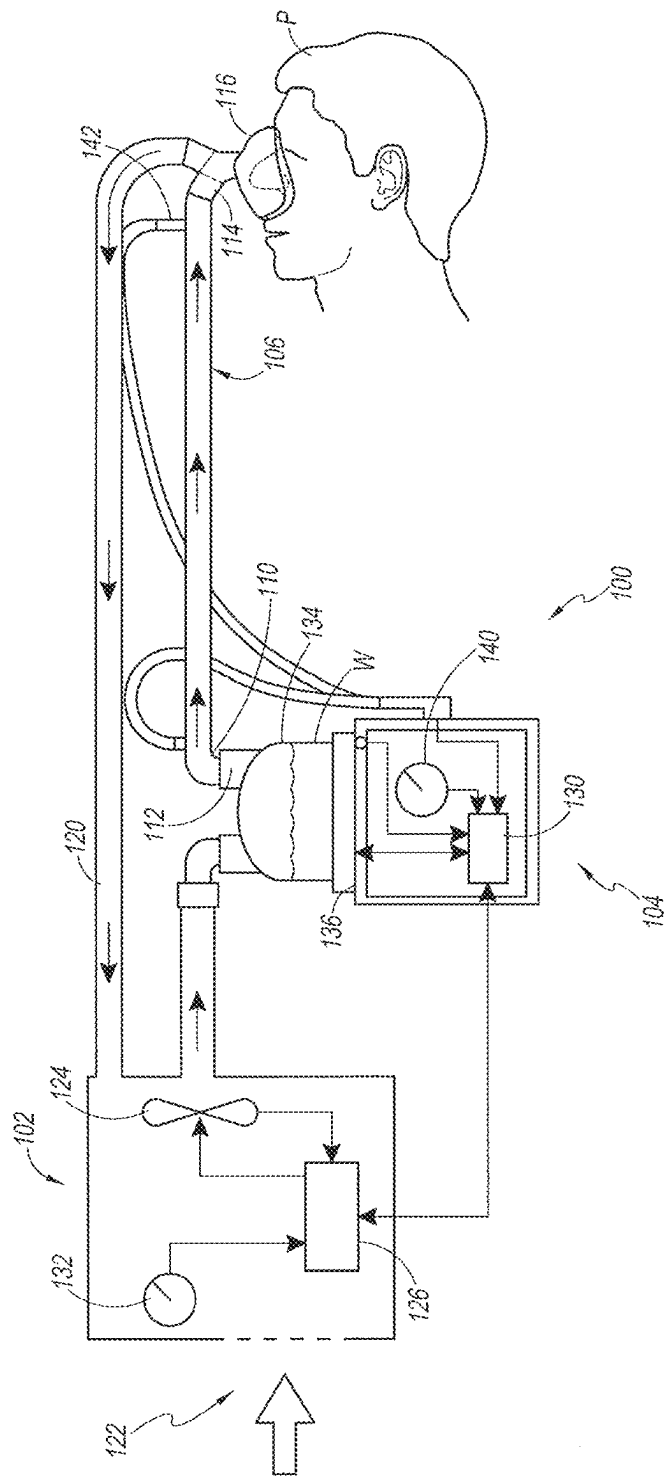


FIG. 1

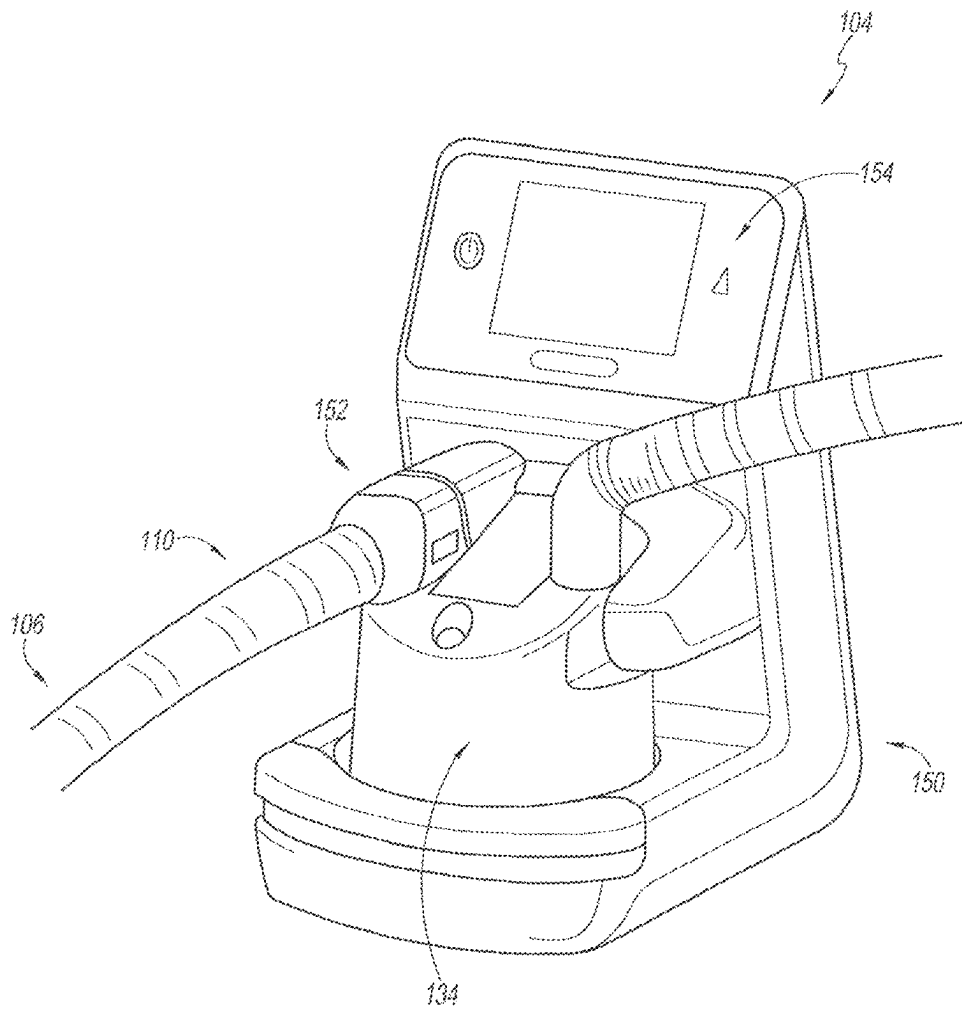


FIG. 2

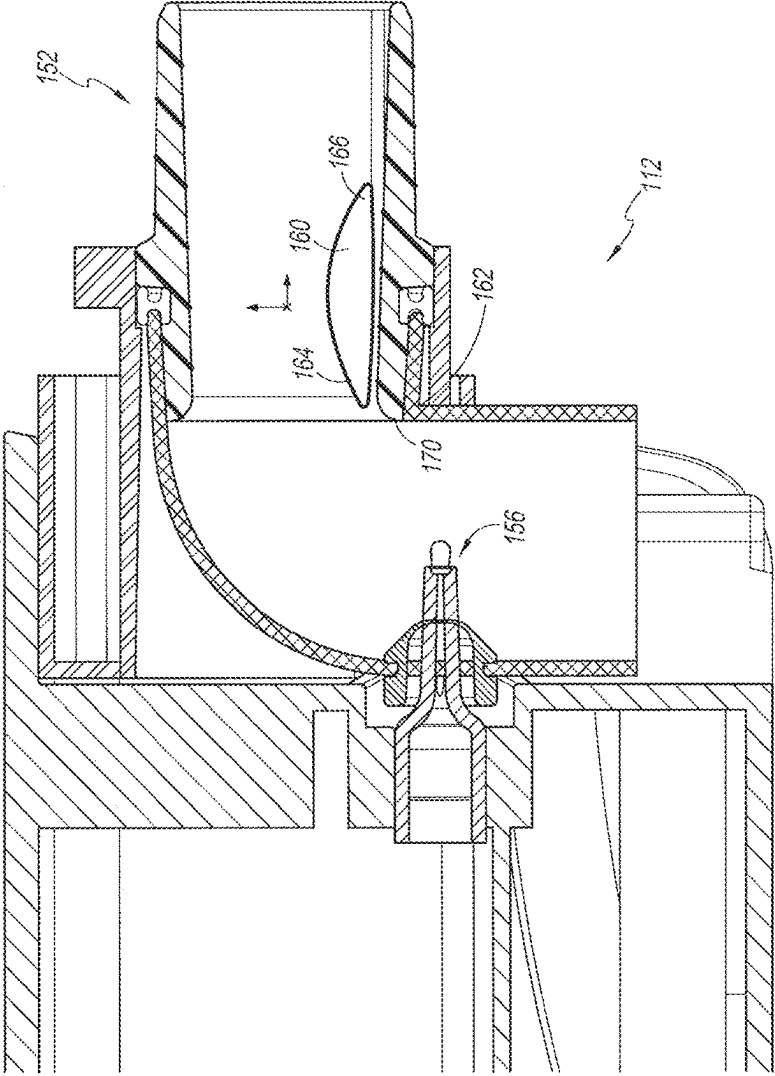


FIG. 3

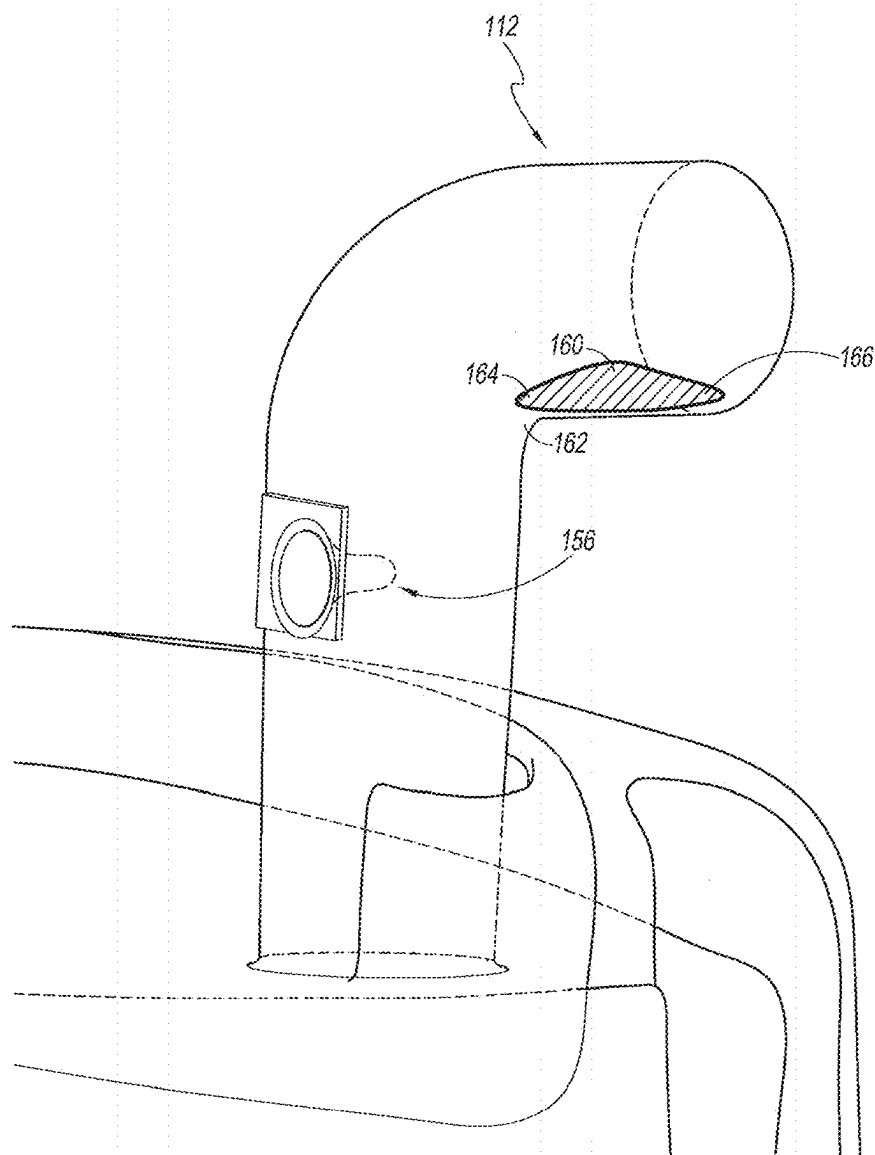


FIG. 4

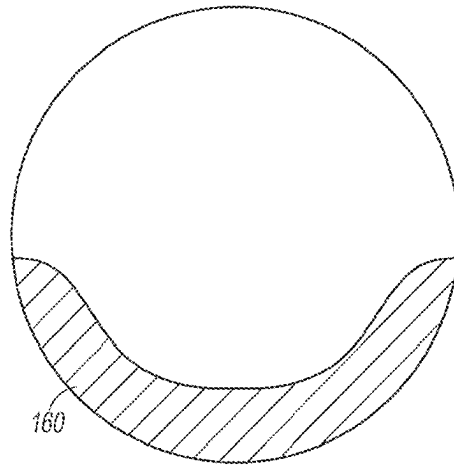


FIG. 5

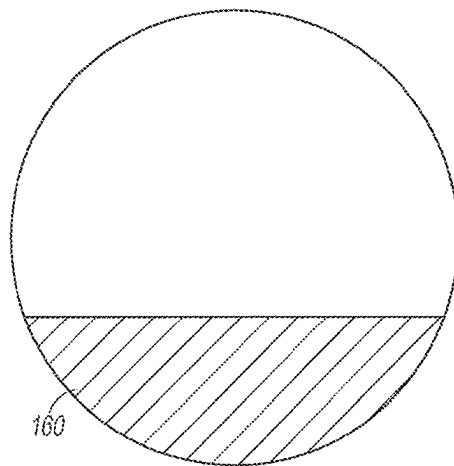


FIG. 6

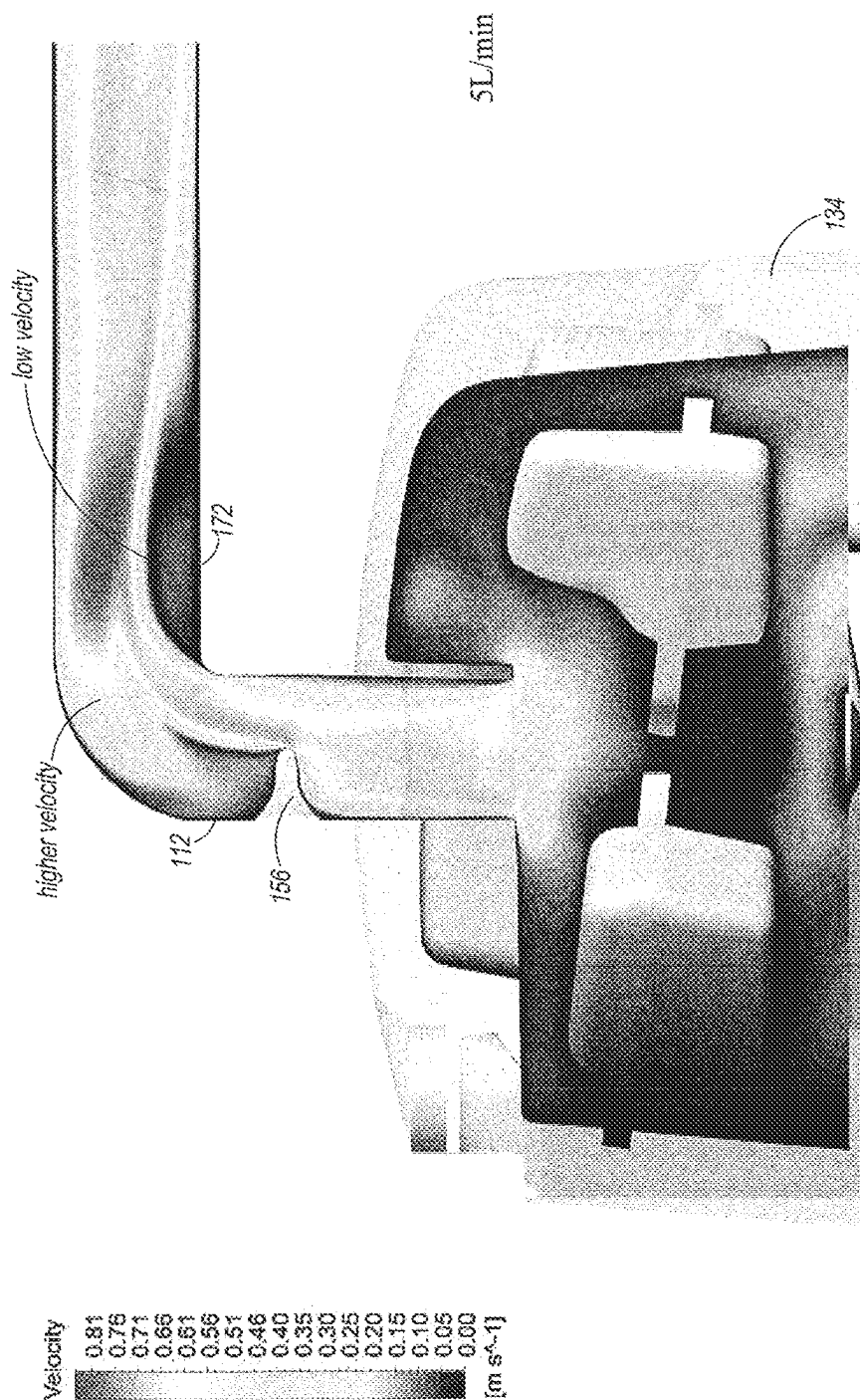


FIG. 7

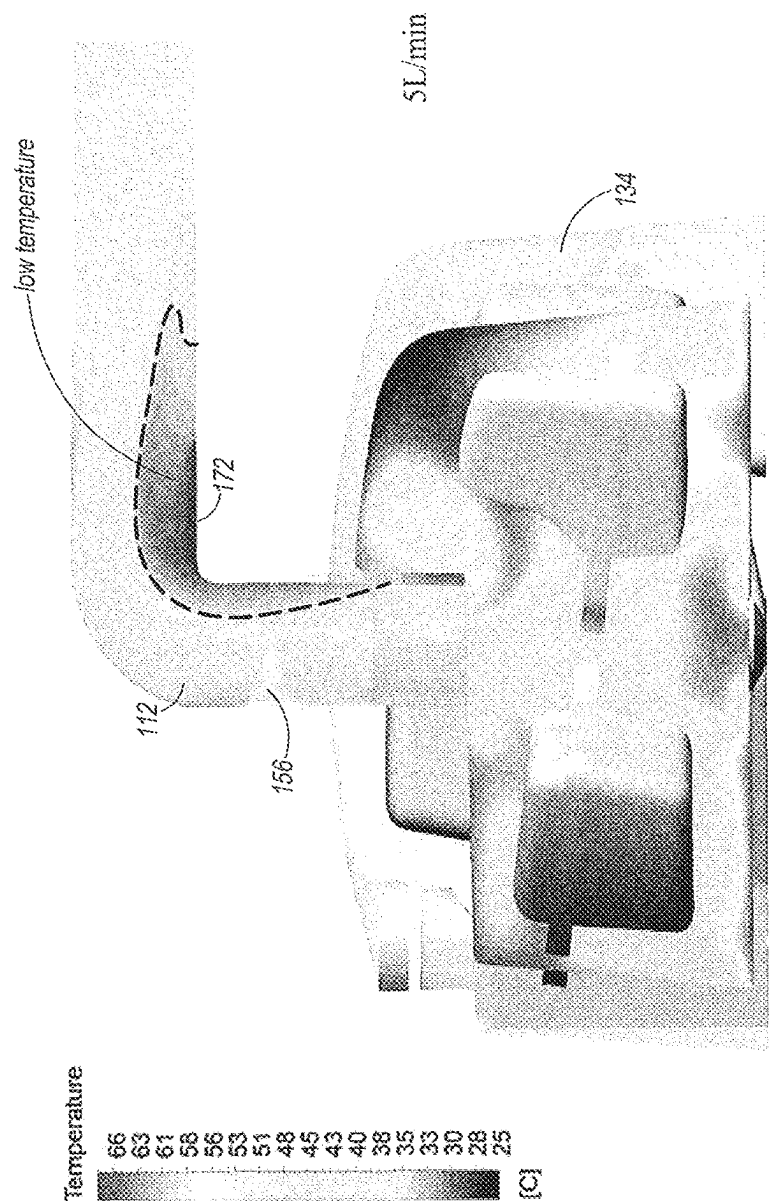


FIG. 8

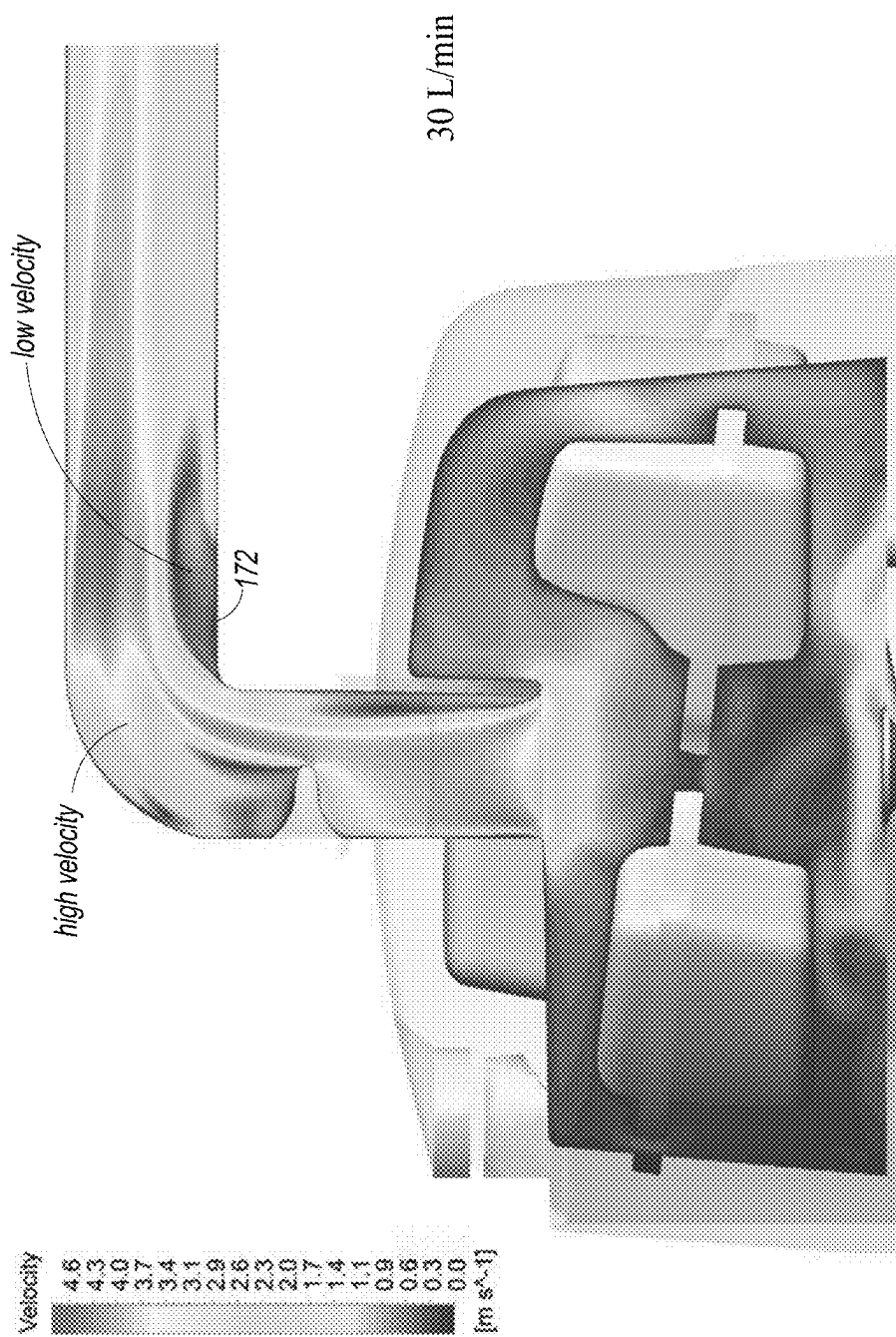


FIG. 9

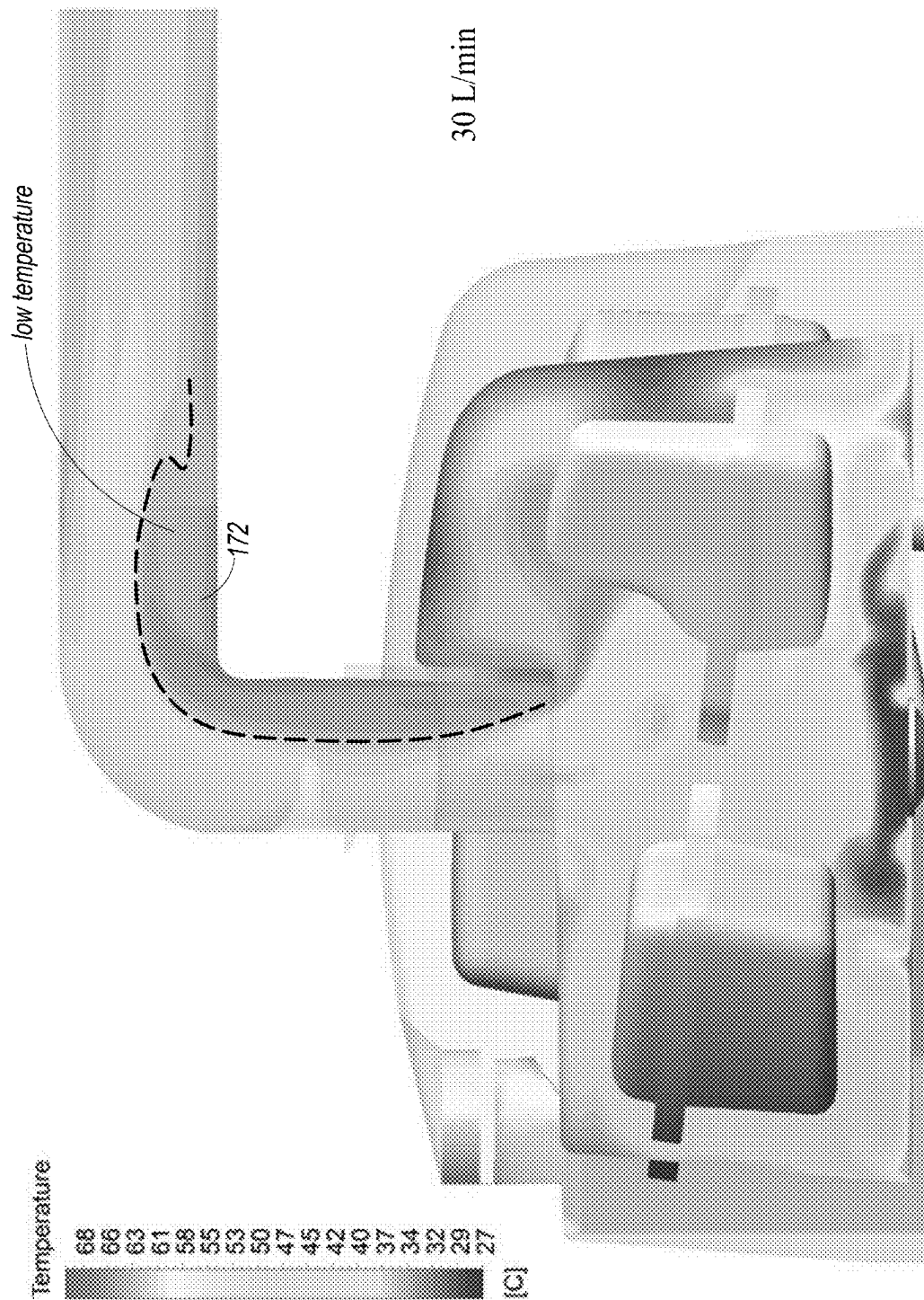


FIG. 10

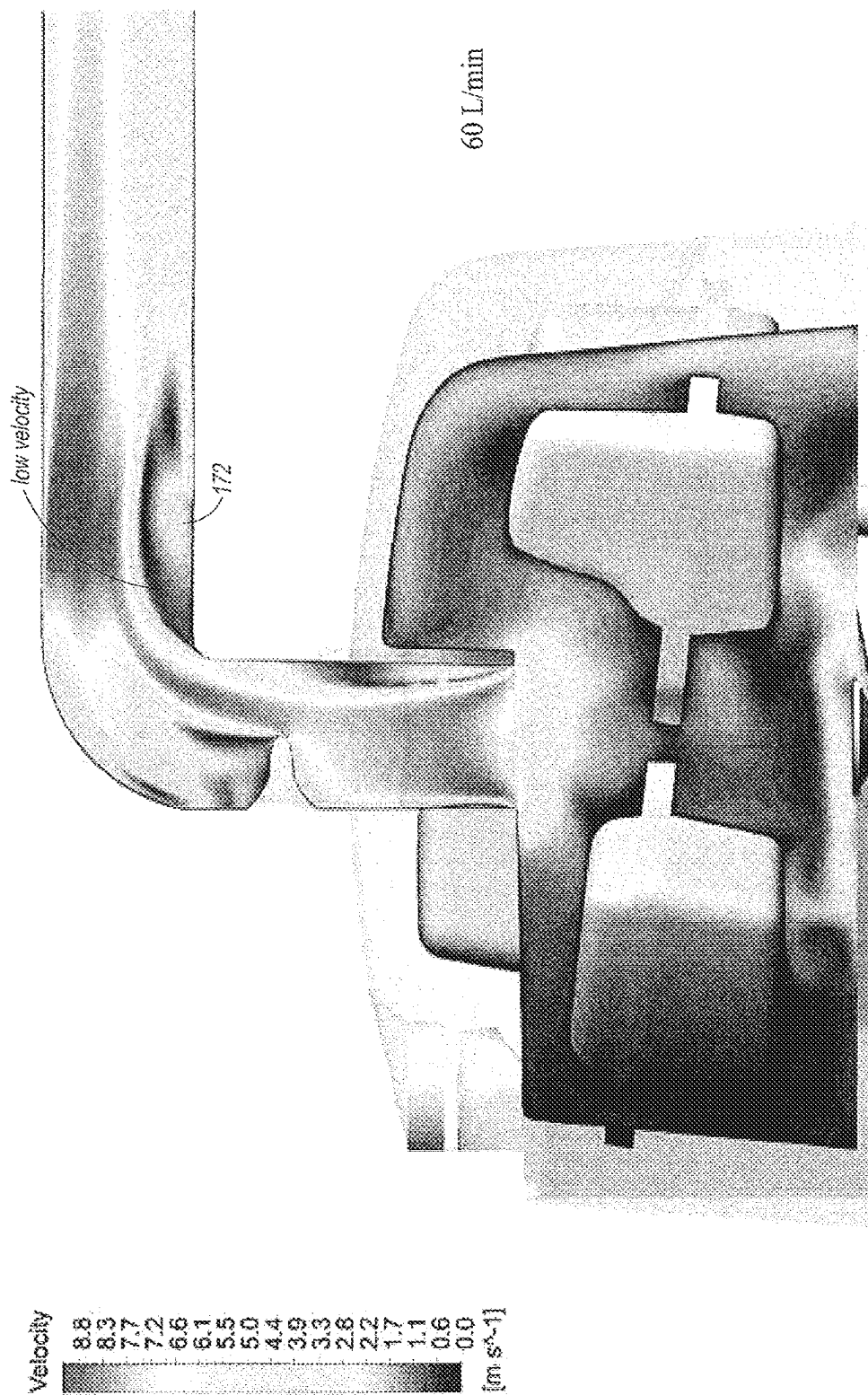


FIG. II

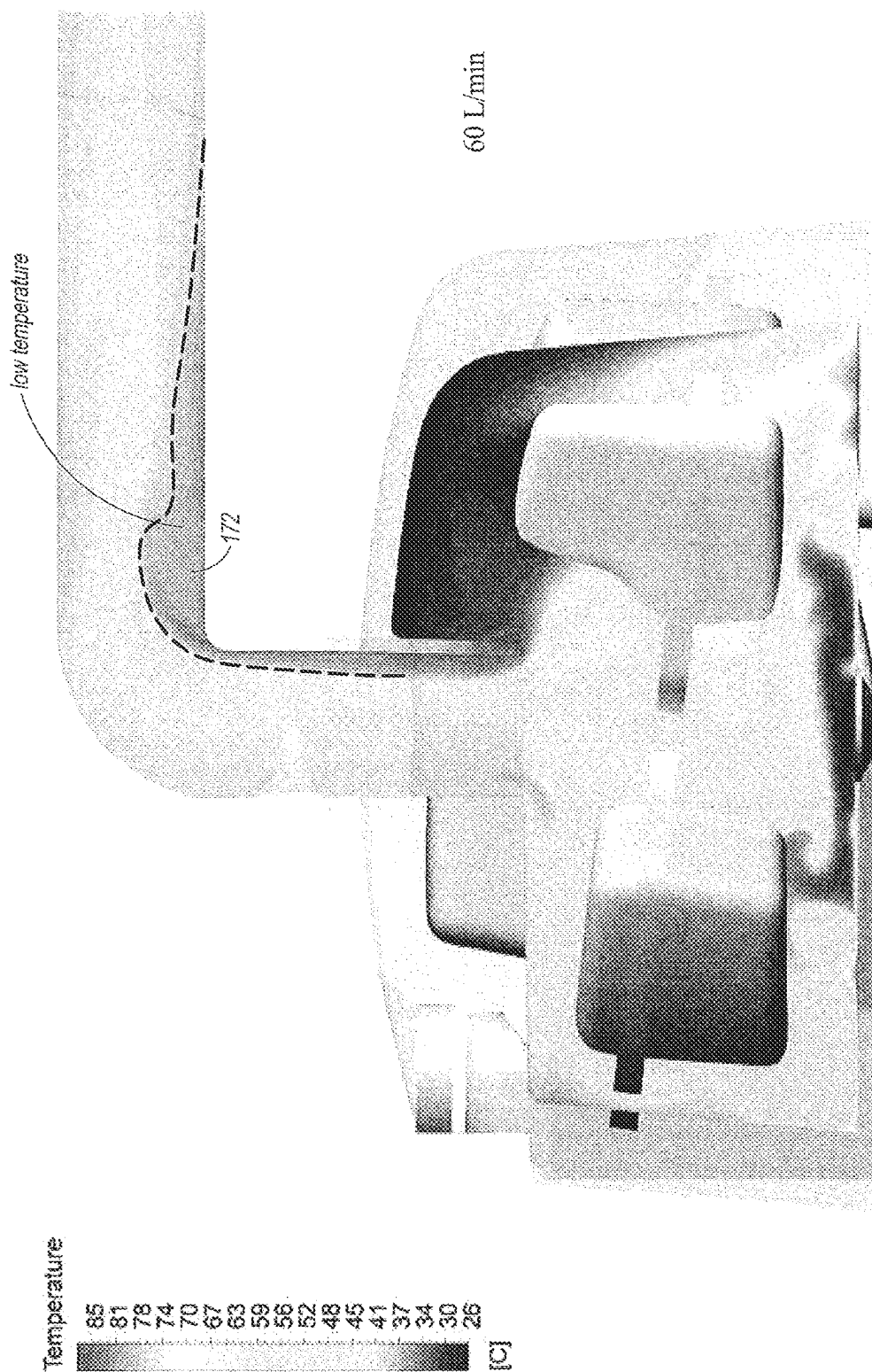


FIG. 12

FIG. 13A

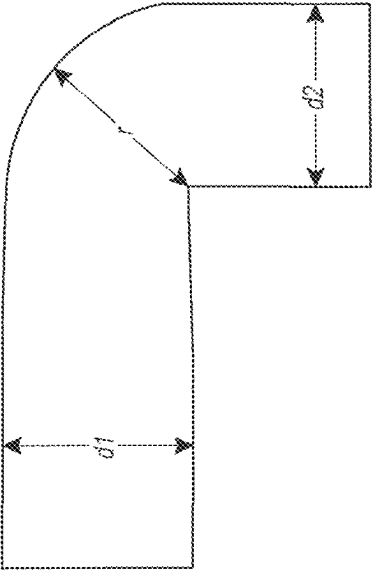


FIG. 13B

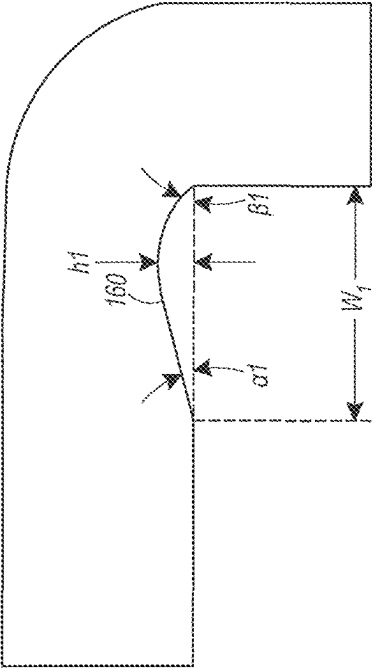


FIG. 13C

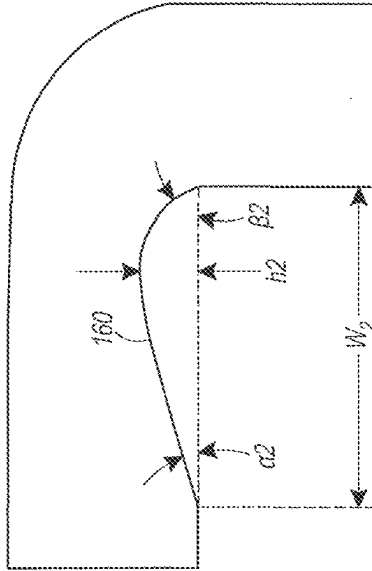


FIG. 13D

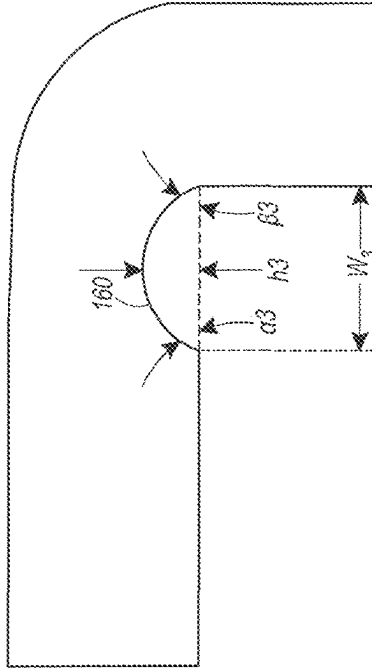


FIG. 14B

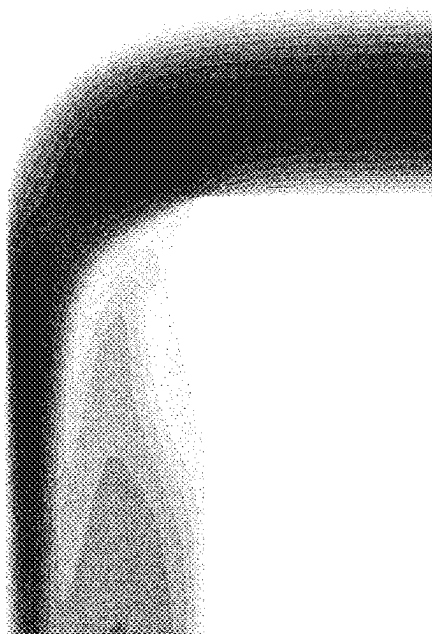


FIG. 14D



FIG. 14A

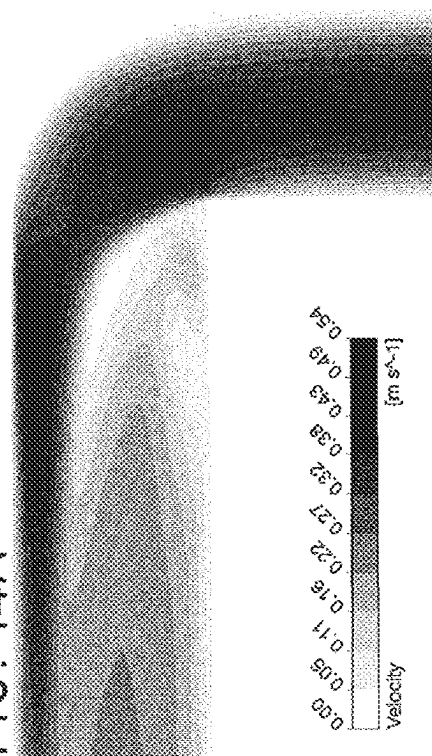


FIG. 14C



FIG. 15B



FIG. 15D



FIG. 15A

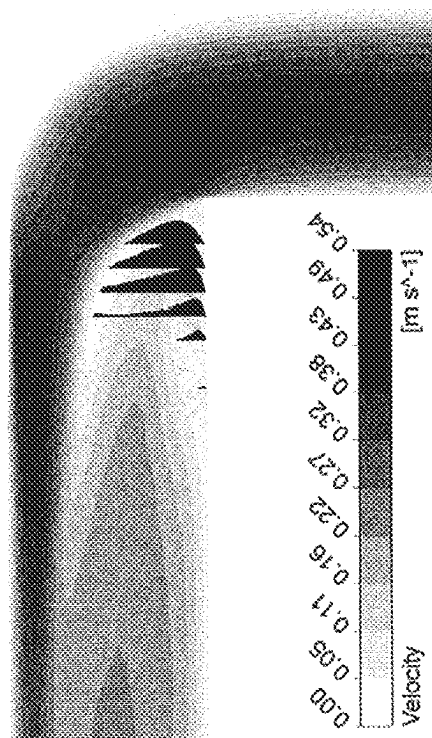
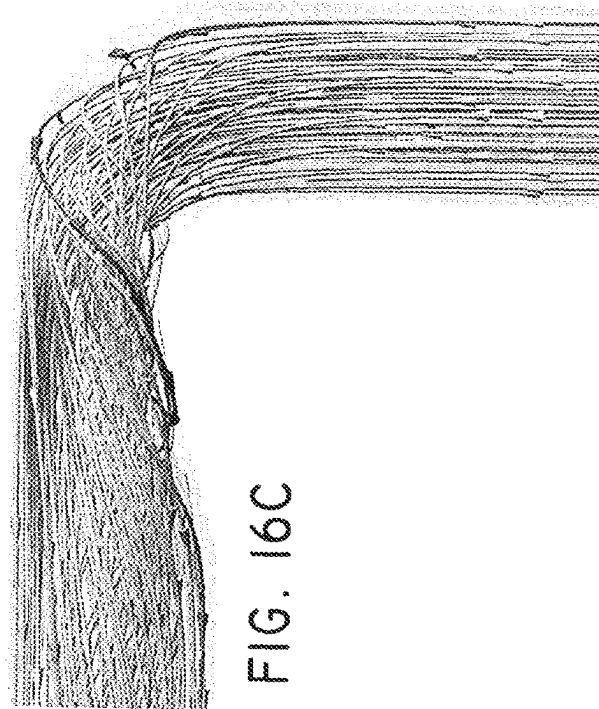
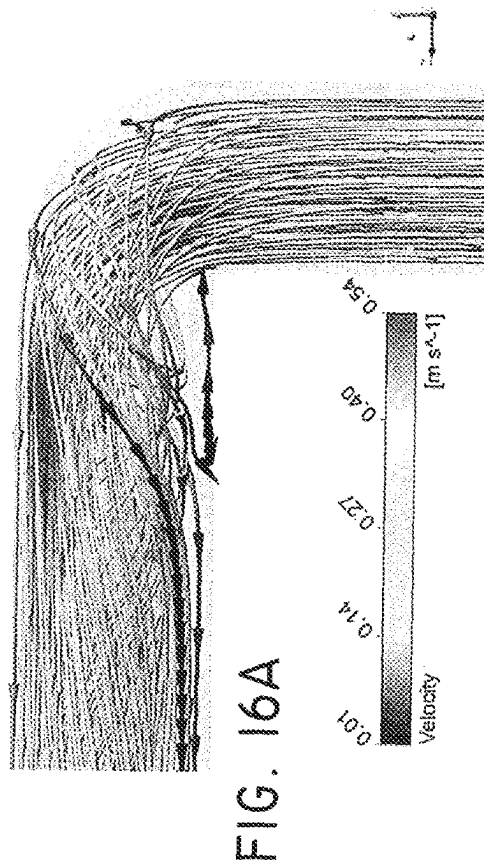
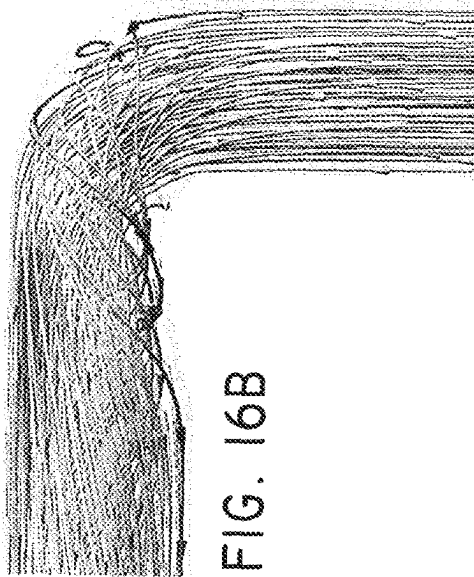


FIG. 15C





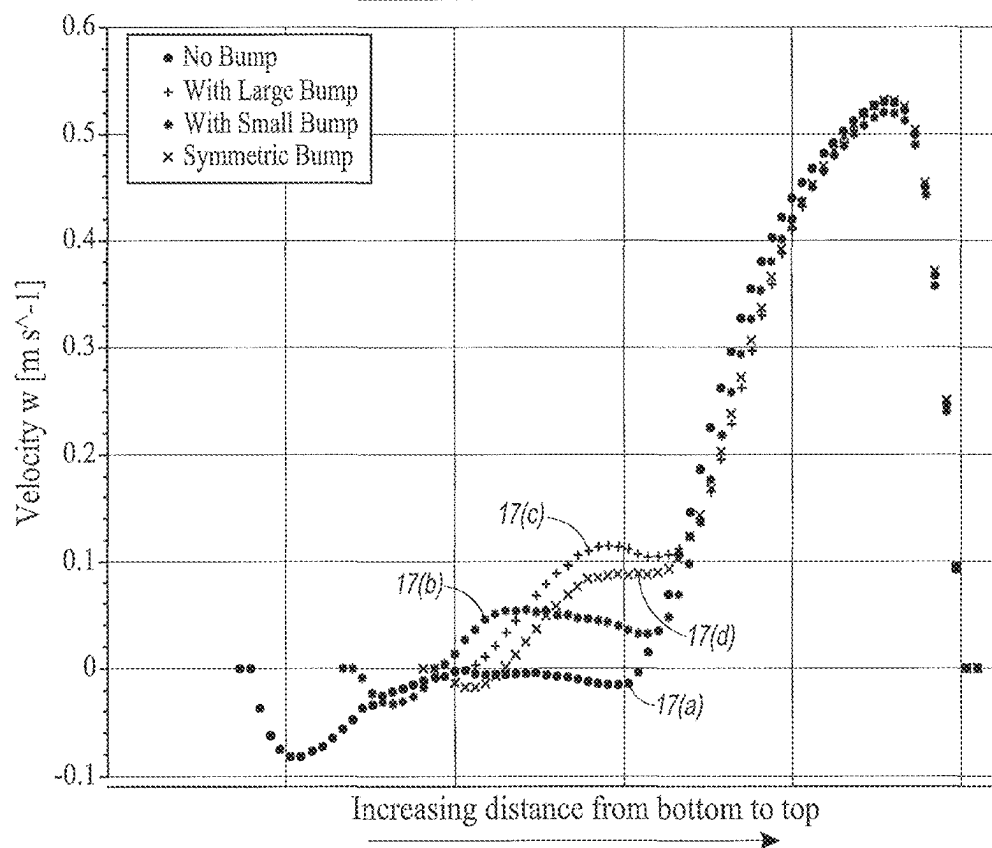
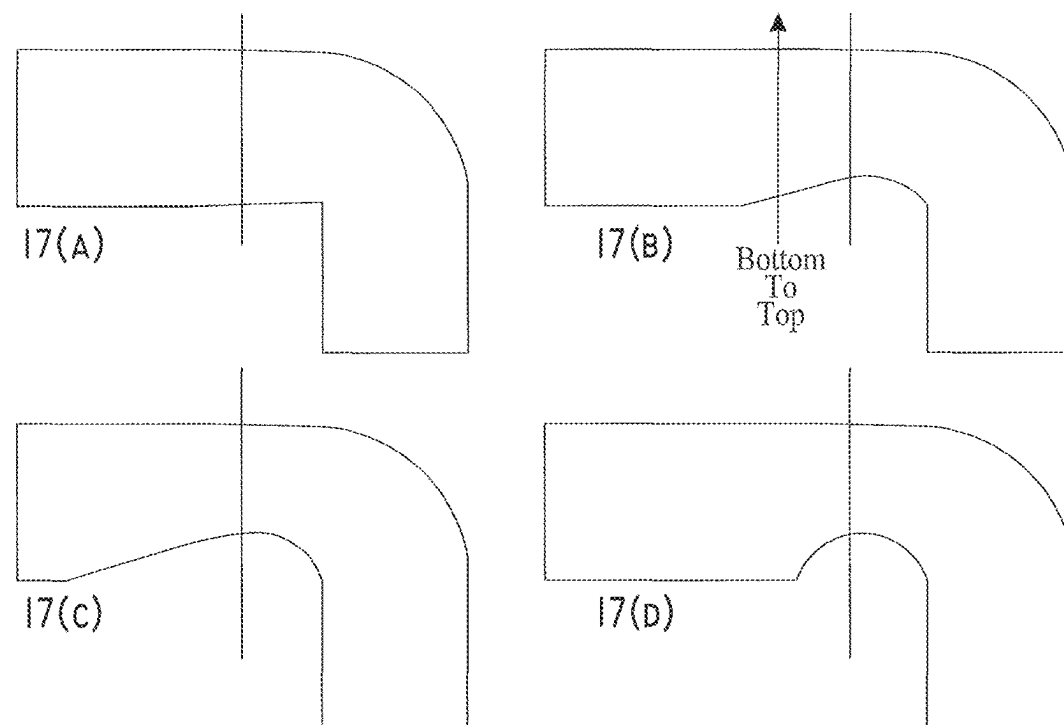


FIG. 17

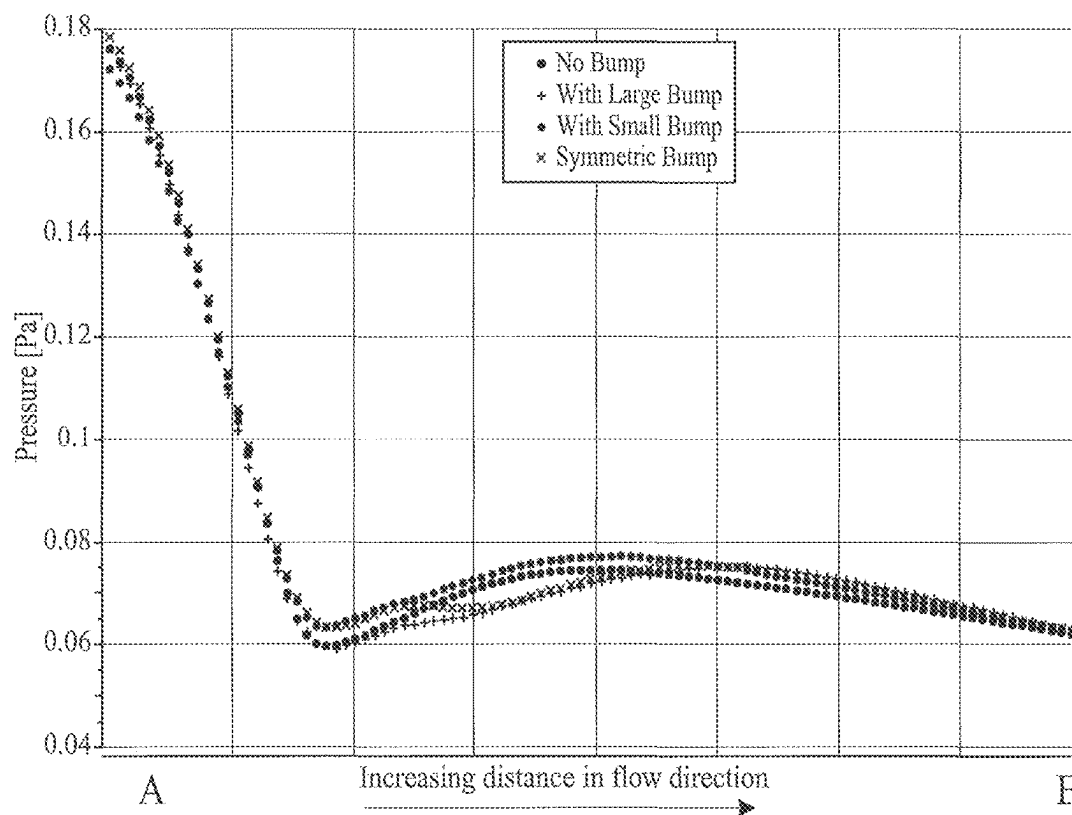
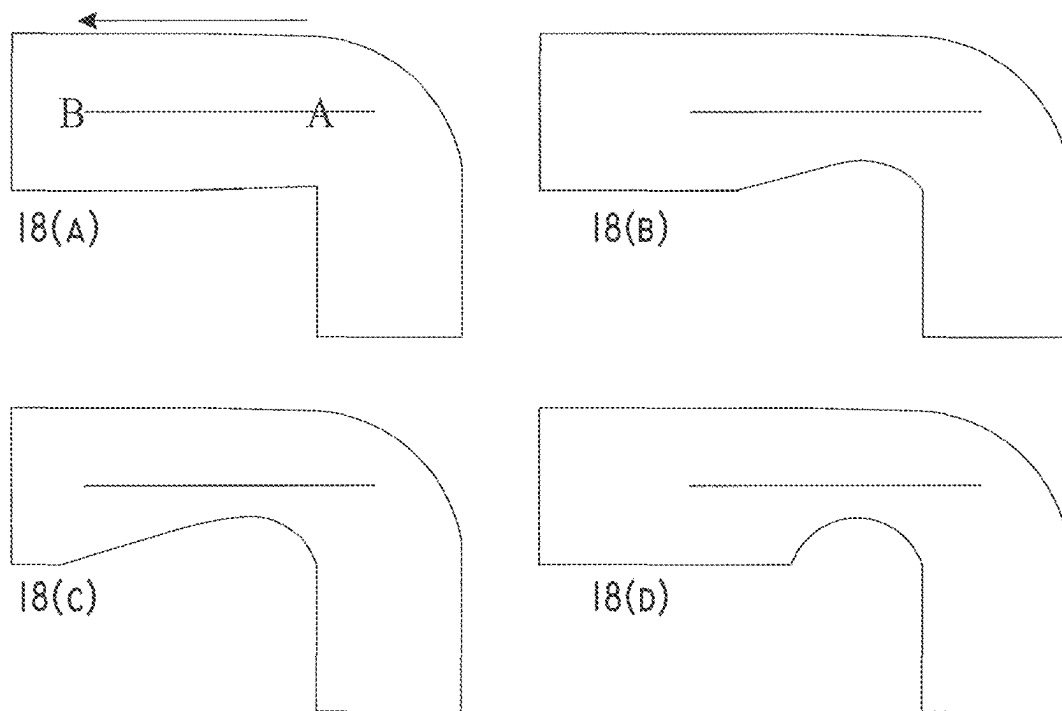


FIG. 18

FIG. 19A

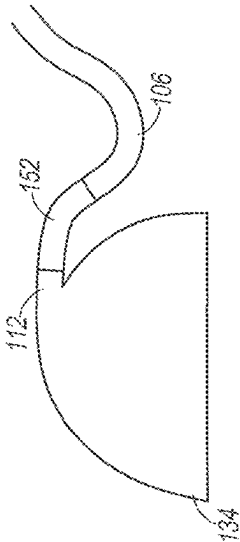


FIG. 19B

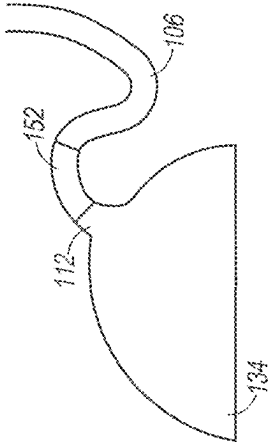


FIG. 19C

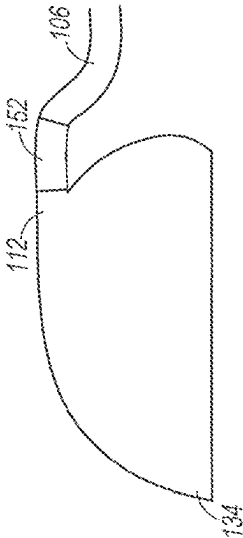
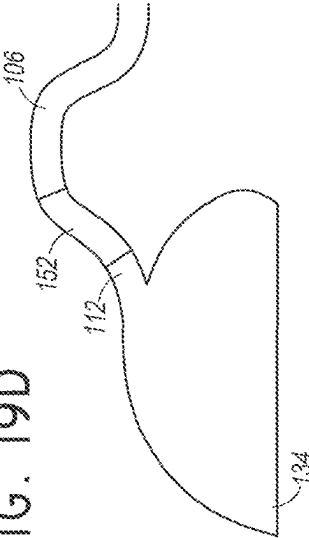


FIG. 19D



RESPIRATORY HUMIDIFICATION SYSTEM**INCORPORATION BY REFERENCE TO ANY
PRIORITY APPLICATIONS**

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND**Technical Field**

The present disclosure generally relates to humidification systems for providing humidified respiratory gases to patients. More particularly, the present disclosure relates to features that improve the performance of humidification systems.

Description of the Related Art

Respiratory humidification systems deliver heated and humidified gases to a patient via a patient interface. Patient interfaces can include a face mask, oral mask, nasal mask, nasal cannula, a combination of oral and nasal mask, tracheal mask, or the like. Gases leaving the humidification chamber to be delivered to a patient are at a temperature and relative humidity that may mimic the transformation of air that occurs as it passes through the nose to the lungs. This promotes efficient gas exchange and ventilation in the lungs, which aids defence mechanisms in the airways and increases patient comfort during treatment. Condensation within the humidification system can occur when gases with high humidity come into contact with a component having cooler walls.

SUMMARY

Delivery of humidified gases to patients exists in the prior art, however an aspect of at least one of the embodiments disclosed herein includes the realisation that there are problems with the delivery of this humidified gas. Horizontal connectors that connect between a humidification chamber and a conduit may allow condensate from the conduit or patient interface to run back into the humidification chamber in use. Elbow ports can be used to form a horizontal connection between a humidification chamber and a conduit, which can cause recirculation to occur resulting in stagnant zones, which cause greater heat loss to occur in these regions. This results in regions of low temperatures, which are more likely to form condensate. Condensate may also form at the connector, as this is an unheated section of the gas pathway. Condensate at the connector may form at the inner surface of the outer wall, before pooling at the base of the conduit, where it may flow back to the humidification chamber.

A system is disclosed which may improve the fluid characteristics of the respiratory gas. Thus, it may reduce the amount of condensate. In some configurations, certain aspects can reduce the amount of condensate that flows back to the humidification chamber relative to the prior art. In some configurations, the angle of the elbow port can be increased to greater than 90° to reduce the recirculation that causes stagnant zones to occur and thereby reduce the condensate that forms in the system. Some embodiments include an insert or other structure shaped as a raised

portion, which can be located near the sharp corner of the elbow port, to fill at least some of the dead space that is formed due to the recirculation of gases that occurs. The insert may reduce the likelihood of recirculation occurring in this region and thus may reduce the likelihood of stagnant zones forming. Therefore, regions of low temperature and low velocity gas are reduced, which reduces the amount of condensate that is generated. A raised portion may also act as barrier to any condensate that is formed and, as such, reduce the likelihood of condensate flowing back to the humidification chamber. A raised portion may be added to the port of the humidification chamber, may be part of the connector that inserts when the connector is connected to the port, or may be part of the conduit.

Some embodiments may comprise either the humidification chamber outlet port or the conduit being shaped in such a way that, or including geometries such that, gravity may reduce the likelihood of condensate reaching the humidification chamber.

Therefore, the improved system may reduce condensate formation between the outlet port of the humidification chamber and the proximal conduit end by improving flow characteristics. The improved system may provide mechanisms to reduce the likelihood of condensate flowing back to the humidification chamber.

In some configurations, a respiratory humidification system comprises a humidification chamber that comprises an outlet port; a conduit that comprises a connector configured to engage with the outlet port such that the conduit forms a gases flow path from the humidification chamber; and a raised portion located in the gases flow path directly adjacent to a sharp corner disposed along the gases flow path within a region defined between the body of the humidification chamber and the conduit.

In some such configurations, the raised portion inhibits liquid flowing from the conduit to the humidification chamber in use.

In some such configurations, the raised portion fills or at least partially fills a dead space region in the gases flow path in use.

In some such configurations, the outlet port comprises a vertical section and a horizontal section and the connector is configured to engage with the horizontal section.

In some such configurations, the raised portion is located within the horizontal section of the outlet port.

In some such configurations, the outlet port comprises a vertical section, the connector comprises a vertical section and a horizontal section, and the vertical section of the connector is configured to engage with the outlet port.

In some such configurations, the raised portion is located within the horizontal section of the connector.

In some such configurations, the raised portion is directly attached to the outlet port.

In some such configurations, the raised portion is directly attached to the connector.

In some such configurations, the raised portion is attached to the conduit.

In some such configurations, the conduit comprises one or more heating elements.

In some configurations, a respiratory humidification system comprises a humidification chamber configured to contain a volume of liquid. The humidification chamber has an outlet port. A conduit is connectable to the outlet port of the humidification chamber. The conduit and the outlet port of the humidification chamber are connected by a connector. A gases flow path is defined from an entrance to the outlet port of the humidification chamber to an outlet of the conduit. A

3

sharp corner is positioned along the gases flow path at a location where the gases flow path makes an abrupt change in direction. A raised portion is located in the gases flow path directly adjacent to the sharp corner.

In some such configurations, the raised portion is located immediately downstream from the sharp corner.

In some such configurations, the raised portion is located within a region of the gases flow path that would be a recirculation region without the raised portion present.

In some such configurations, the raised portion forms a portion of the outlet port.

In some such configurations, the raised portion forms a portion of the connector.

In some such configurations, the raised portion forms a portion of the conduit.

In some such configurations, the raised portion fills at least a lower portion of at least one of a horizontal portion of the outlet port, a conduit or a connector such that condensate is less likely to flow from the conduit into the humidification chamber.

In some such configurations, the raised portion has a tapered edge.

In some such configurations, the raised portion has a straight edge.

In some such configurations, the raised portion is asymmetrical along its length.

In some such configurations, the raised portion is symmetrical along its length.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects, and advantages of the present disclosure will be described with respect to the following figures, which are intended to illustrate and not to limit the preferred embodiments.

FIG. 1 is a schematic view of an example embodiment of a patient gas supply system.

FIG. 2 is a perspective view of a humidifier useable in the system of FIG. 1.

FIG. 3 is an enlarged section view of a portion of the humidifier of FIG. 2 and showing a first configuration of a raised portion within the gases flow path.

FIG. 4 is a view of a portion of a humidifier chamber useable with the humidifier of FIG. 2 and showing a second configuration of a raised portion within the gases flow path.

FIGS. 5 and 6 show cross-sections of raised portions.

FIG. 7 shows a velocity profile of a humidification chamber with an elbow port for a 5 L/min flow.

FIG. 8 shows a temperature profile of a humidification chamber with an elbow port for a 5 L/min flow.

FIG. 9 shows a velocity profile of a humidification chamber with an elbow port for a 30 L/min flow.

FIG. 10 shows a temperature profile of a humidification chamber with an elbow port for a 30 L/min flow.

FIG. 11 shows a velocity profile of a humidification chamber with an elbow port for a 60 L/min flow.

FIG. 12 shows a temperature profile of a humidification chamber with an elbow port for a 60 L/min flow.

FIGS. 13(a)-13(d) show different configurations of gas flow paths with three different raised portions illustrated in FIGS. 13(b)-13(d).

FIGS. 14(a)-14(d) illustrate flow velocity contours through the configurations of FIGS. 13(a)-13(d) for a 5 L/min flow.

FIGS. 15(a)-15(d) illustrate reverse flow vectors overlaid onto the flow velocity contours of FIGS. 14(a)-14(d).

4

FIGS. 16(a)-16(c) illustrate show streamlines coloured by velocity magnitude and indicating paths taken by a zero mass particle if that particle were traveling with the flow.

FIG. 17 is a plot illustrating the flow velocity components in a 5 LPM flow relative to the direction of flow at the same cross section for each of the configurations of FIGS. 13(a)-13(d).

FIG. 18 is a plot comparing pressure magnitudes taken along a horizontal line that extends along an axis of the configurations of FIGS. 13(a)-13(d).

FIGS. 19(a)-19(d) show additional humidification chamber embodiments designed to impede condensate flowing back into the humidification chamber.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

With reference to FIG. 1, a system **100** is illustrated that can be used to supply heated and/or humidified gases flow to a patient or other user. The system **100** can be configured to be a continuous, variable, or bi-level positive airway pressure (PAP) system, invasive or non-invasive respiratory assistance system, high flow respiratory therapy system, a surgical insufflation system, or a system providing other forms of medical gases. The system **100** may be readily adapted for other applications involving the supply of a heated and/or humidified gas flow to a user or patient, including but not limited to laparoscopy, ventilation support, and the like. Such applications may use alternative gases, operating parameters (e.g., flow, pressure, temperature, or humidity) and patient interfaces.

In the system **100**, dry or relatively dry gases pass from a gases source **102** to a humidifier **104**. The gases source **102** may be, for example, a ventilator or a blower.

The humidifier **104** conditions the dry or relatively dry gases. For example, the humidifier **104** can supply heat or humidity to the dry or relatively dry gases.

An inspiratory tube **106** is used to deliver the conditioned gases to a patient P. In the illustrated configuration, a distal end **110** of the inspiratory tube **106** connects to a port **112** of the humidifier **104**. Thus, the humidifier **104** supplies conditioned gases to the inspiratory tube **106**.

In the illustrated configuration, the conditioned gases flow through the inspiratory tube **106** to a Y-piece **114**. A patient interface **116** (e.g., a mask) receives the conditioned gases from the Y-piece **114** and the patient interface **116** supplies the conditioned gases to the patient P. Any suitable patient interface may be incorporated. Patient interface is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art (that is, it is not to be limited to a special or customized meaning) and includes, without limitation, masks (such as tracheal mask, face masks, oral masks, nasal masks, a combination of oral and nasal mask, for example), cannulas (nasal cannulas, for example), and nasal pillows.

In the illustrated configuration, an expiratory tube **120** also connects to the patient interface **116** through the Y-piece **114**. The expiratory tube **120** may be configured to move exhaled gases away from the patient P. In the illustrated configuration, the expiratory tube **120** returns exhaled gases from the patient P to the gases source **102**.

In illustrated configuration, dry or relatively dry gases enter the gases source **102** through a vent **122** or other inlet. A fan, blower or other flow generator **124** may improve gas flow into the gases source **102** by drawing air or other gases through the vent **122**. The flow generator **124** may be, for instance, a variable speed fan.

5

An electronic controller **126** controls the speed of the flow generator **124**. In particular, the function of the electronic controller **126** may be controlled by an electronic master controller **130** in response to inputs to the master controller **130** and a user-set predetermined required value (e.g., a preset value) of pressure or fan speed. The value can be set using an input component, such as a dial **132**, for example but without limitation.

The humidifier **104** comprises a humidification chamber **134**. The humidification chamber **134** comprises the port **112**. The body of the humidification chamber **134** can contain a volume of water **W** or other suitable humidifying liquid. The humidification chamber **134** is removable from the humidifier **104** after use to allow the humidification chamber **134** to be more readily sterilized or disposed.

The body of the humidification chamber **134** may be formed from a non-conductive glass or plastics material. The humidification chamber **134** may comprise conductive components. For instance, the humidification chamber **134** may comprise a highly heat-conductive base (for example, an aluminum base). The heat-conductive base can contact or associate with a heater plate **136** on the humidifier **104**.

The humidifier **104** may also include electronic controls. In the illustrated configuration, the humidifier **104** includes the master controller **130**. The master controller **130** can be an electronic, analog, or digital master controller. The master controller **130** may be a microprocessor-based controller executing computer software commands stored in associated memory. In response to humidity or temperature values provided via a user interface **140**, for example, and other inputs, the master controller **130** determines when (or to what level) to energize the heater plate **136** to heat the water **W** within the humidification chamber **134**.

A sensor probe **142** may connect to the inspiratory tube **106** near the Y-piece **114**, or directly to the Y-piece **114** or the patient interface **116**. The sensor probe **142** monitors the temperature of the flow of gases near or at the patient interface **116**. A heating filament (not shown) may be used to adjust the temperature of the patient interface **116**, the Y-piece **114**, and/or the inspiratory tube **106** to raise or maintain the temperature of the flow of gases above the saturation temperature, thereby reducing the opportunity for unwanted condensation.

In FIG. 1, exhaled gases are returned from the patient interface **116** to the gases source **102** via the expiratory tube **120**. The expiratory tube **120** may have a sensor probe and/or heating filament, as described above with respect to the inspiratory tube **106**, integrated with it to reduce the opportunity for condensation. Furthermore, the expiratory tube **120** need not return exhaled gases to the gases source **102**. Alternatively, exhaled gases may be passed directly to ambient surroundings or to other ancillary equipment, such as an air scrubber/filter (not shown). In certain embodiments, the expiratory tube **120** is omitted altogether.

With reference to FIG. 2, a humidification apparatus **150** that can be used as the humidifier **104** in the system **100** is shown. As illustrated, a connector **152** pneumatically connects the distal end **110** of the inspiratory tube **106** to the port (not shown) of the humidification chamber **134**. The connector **152** may also facilitate electrical connection to the humidifier **104**. The conduit **106** may comprise one or more resistive heating wires that provide for heating of gases flowing through the conduit **106** and/or sensor wires that electrically or otherwise facilitate communication of signals relating to one or more parameters of the system **100**. Thus,

6

the term “electrical connection” is used to distinguish from “pneumatic connection” and should not be construed in a limiting way.

The humidifier **104** further includes a panel **154**. The panel **154** may be used to mount a user display and/or controls. For example, various dials, switches, and other input means may be used to control operation of the device. Additionally or alternatively, a touch screen display may be used. The user display may display parameters of the system **100**, warnings in the event of any errors or malfunctions, or prompts where user action is required, etc. Where a touch screen display is used, the same display may be used to present information to a user and receive inputs from a user, at least in part.

As illustrated in FIG. 2, the conduit **106** is mounted to the humidification chamber **134** such that the conduit **106** extends substantially parallel to the direction of motion of the humidification chamber **134** as it is slid on or off of the humidifier **104**. With reference to FIG. 3, a sectioned view of a portion of the humidifier **104** is shown. As shown, a probe **156** can be positioned within the outlet port **112** upstream of the connector **152** that connects the chamber **134** to the inspiratory tube **106**. In some configurations, the probe **156** can be positioned in the port **112**. In some configurations, the probe **156** can be positioned within the connector **152**. In some configurations, the port **112** is formed as an elbow with a vertically extending portion and a horizontally extending portion. Other configurations are possible.

In configurations with the elbow port **112** having a sharp corner, it has been discovered that a significant amount of recirculation occurs in the horizontal portion of the port **112**. More particularly, with reference to FIGS. 7-12, a sharp corner at the inside lower transition from vertical to horizontal causes a significant amount of recirculation within the flow. This recirculation, in effect, creates a significant region of dead space. The gases flow can cool within the recirculation/dead space region, which can cause condensation to occur within this region.

FIGS. 7-12 show examples of the velocity and temperature profiles that can be generated for flow rates of 5 L/min, 30 L/min and 60 L/min respectively. The system **100** is not limited to these flow rates; they merely provide an indication of the relationship between different flow rates and the generated fluid characteristics. It appears that a larger dead space region may be generated at higher flow rates.

The system **100** disclosed herein may improve the fluid characteristics of the respiratory gas and may reduce the amount of condensate or other fluid that flows back to the humidification chamber **134** from the attached conduit **106**. To improve the fluid characteristics in the system **100**, the angle of the elbow port **112** of the humidification chamber **134** may be increased; this may soften the sharp corner that is shown in FIGS. 7-12.

With reference again to FIG. 3, a raised portion (or insert) **160** can be attached to, or integrally formed with, one or more components of the system **100**. For example, the raised portion **160** is shown in FIG. 3 as being positioned within the connector **152**. By way of another example, the raised portion **160** is shown in FIG. 4 as being positioned within the port **112**. In some configurations, the raised portion **160** can be positioned within the inspiratory conduit **106** or the like. The raised portion **160** may be inserted or added to a component using moulding techniques, adhesives, or other appropriate ways of attachment.

In some configurations, an elbow portion **162** can be defined at the transition from a first direction to a second

direction of the port 112. In some configurations, the elbow portion 162 is defined by a transition from a first direction to a second direction that is generally normal to the first direction. In some configurations, the elbow portion 162 is defined by a transition from vertical to horizontal.

As shown in FIG. 3 and FIG. 4, the raised portion 160 can be positioned directly adjacent to the elbow portion 162. As used herein, directly adjacent has its ordinary meaning of sharing a border or boundary. In some configurations, the raised portion 160 is positioned within the gas flow of only one of the two portions of the port 112. In other words, in the illustrated configurations, the raised portion 160 protrudes into the gas flow path of the horizontal portion but does not protrude into the gas flow path of the vertical portion. In some configurations, at least the upstream end 164 of the raised portion 160 is positioned entirely within a region that would be a recirculation region without the raised portion 160 present. In some such configurations, both the upstream end 164 and the downstream end 166 of the raised portion 160 are positioned within a region that would be a recirculation region without the raised portion 160 present. In some configurations, the entire raised portion 160 is positioned within a region that would be a recirculation region without the raised portion 160 present. In some configurations, the raised portion 160 is located in the gases flow path directly adjacent to a sharp corner 170 created by the angle of the elbow portion 162 disposed along the gases flow path within a region defined between the humidification chamber 134 and the conduit 106. In some configurations, the sharp corner 170 is positioned along the gases flow path at a location where the gases flow path makes an abrupt change in direction.

With reference now to FIG. 5 and FIG. 6, the raised portion 160 may resemble a speed bump, protrusion, insert, or the like. The raised portion 160 can be integrally formed with the port 112, the connector 152, or another component that joins the humidification chamber 134 and the inspiratory conduit 106. The raised portion 160 can be a separate and/or separable component from the port 112, the connector 152, or other component and secured thereto in any suitable manner.

As shown in FIG. 4, the raised portion 160 is attached to the outlet port 112 of the humidification chamber 134. The outlet port 112 comprises the elbow portion 162, which, in this case, creates an angle of, but not limited to, 90°. The raised portion 160 is located at or near the sharp corner 170 created by the angle of the elbow portion 162 because the fluid may be more likely to follow the surface of the port 112 if it is a smooth curve.

The size of the raised portion 160 can be determined by the temperature and velocity profiles, examples of which are shown in FIGS. 7-12, which estimate the size of a dead space region 172 present in the system 100 at high flows. As discussed above, the dead space region 172 may appear to be larger at higher flow rates and, therefore, if the raised portion 160 is made to accommodate these flow rates, it may also accommodate lower flow rates. Occlusion of flow is less likely to occur at lower flow rates and, because the raised portion 160 fills the regions of otherwise dead space present in the system 100, there are minimal limitations of flow. Thus, the raised portion 160 may accommodate higher flow rates, although, in some embodiments, lower flow rates may also be accommodated. The system 100 discloses the use of the raised portion 160 to accommodate flows between about 5 L/min and about 60 L/min. Some embodiments may accommodate flow rates that exceed these values, for example between 0 L/min and about 100 L/min.

FIG. 3 shows the raised portion 160 inserted into the port 112 as the connector 152 is connected with the humidification chamber port 112. The raised portion 160 may be attached to the connector 152, the raised portion 160 may be attached to the conduit 106 or, in some embodiments, the raised portion 160 may act as a tongue that may be attached to the connector 152 or the conduit 106 and that may extend past the edge of the connector 152 into the outlet port 112. In some embodiments, the connector 152 may be a male connector that is used with the raised portion 160 attached at the edge of the connector 152. This is not considered to be an exhaustive list of mechanisms to insert the raised portion 160 into the port 112 of the humidification chamber 134, but merely a listing of examples of mechanisms that could be used, recognising that other mechanisms may also be possible.

The raised portion 160 may be located near the sharp corner 170 to soften the sharp corner 170 and to fill at least a portion of, if not the entirety of, the dead space region 172 that is formed (see FIGS. 7-12). Benefits of inserting the raised portion 160 into the port 112 of the humidification chamber 134 as the connector 152 is inserted include the reduced distance between the outlet port 112 of the humidification chamber 134 and the conduit 106. This may be beneficial because the connector 152 itself is unheated, which allows the gas to cool therein, which can cause the formation of condensate as the gas travels between the outlet port 112 of the humidification chamber 134 and the conduit 106. By reducing the distance between the outlet port 112 of the humidification chamber 134 and the heated portion of the conduit 106, the time available for gas cooling may be reduced. As a result, the amount of condensate that is formed may also be reduced.

The raised portion 160 may also act to occlude some or all of any condensate or other liquid flowing back into the humidification chamber 134. By filling at least a portion of the dead space 172, the lower temperature zone of recirculation can be reduced and, thus, it is likely that less condensate is generated due to the reduced flow separation that may be produced. This may improve the temperature profile of the gas over the unheated connector 152 as it moves towards the heated conduit 106, which may result in less temperature loss as it passes through the unheated connector 152.

FIG. 5 and FIG. 6 show possible cross sections of the raised portion 160, such as that discussed above. FIG. 5 shows the raised portion 160 having a tapered edge while FIG. 6 shows the raised portion 160 having a straight edge. Other configurations also are possible.

With reference to FIGS. 13(a)-13(d), several different configurations that have been investigated are illustrated. Note that FIGS. 13(a)-13(d) are depictions of the gases flow path and, as such, the raised portion 160 is not explicitly shown and the conduit 106, the connector 152, and the port 112 also are not explicitly shown. Rather, these illustrations simply reflect the gas flow path defined by these boundary components.

FIG. 13(a) illustrates a configuration without any raised portion. FIG. 13(a), however, is useful to establish a baseline and to explain some of the dimensions used in the following discussion. As illustrated, the flow path has an outlet diameter d1 and an inlet diameter d2. In some configurations, the outlet diameter and the inlet diameter are the same. In some configurations, the outlet diameter d1 is larger than the inlet diameter d2 (i.e., the conduit 106 has a larger bore diameter than the port 112). In the illustrated configurations, the inlet diameter d2 is 20 mm while the outlet diameter d1 is 21 mm.

Also, in the illustrated configurations, the radius r between the inlet portion and the outlet portion is 10 mm. In each of the configurations shown in FIGS. 13(a)-13(d), these basic dimensions are the same.

FIG. 13(b) illustrates a smaller asymmetric embodiment of the raised portion 160. FIG. 13(c) illustrates a larger asymmetric embodiment of the raised portion 160. FIG. 13(d) illustrates a symmetric embodiment of the raised portion 160. With reference to FIGS. 13(b), 13(c), and 13(d), each of the raised portions 160 can be generally defined in section with the illustrated dimensions. Each of the raised portions 160 will have a first angle (α) at the downstream end 166, a second angle (β) at the upstream end 164, a length (w) and a height (h). A volume reduction in the flow path in the region of the raised portions 160 can be defined as the volume of the raised portion 160 divided by the volume of the cylinder bounded by the length (w) of the raised portion 160 and the outlet diameter ($d1$).

In some configurations, the first angle (α) can be determined once the other dimensions have been determined. For example, in some configurations, the height (h) can be between 15% and 30% of the outlet diameter $d1$. The second angle (β) can be between 30° and 60°. The length (w) can be larger than 18 mm. Thus, the upper limit of the length (w) and the corresponding range for the first angle (α) can be a function of the other values. In some configurations, however, the first angle (α) can be between 50° and 70°. In some configurations, the first angle (α) can exceed 60° such that condensate is more likely to be contained within the conduit 106 rather than flowing back into the humidification chamber 134.

With reference now to FIG. 13(b), the height (h) of the illustrated raised portion 160 is 4 mm. The length (w) of the raised portion 160 is 25 mm. The first angle (α) of the raised portion 160 is 15 degrees and the second angle (β) is 35 degrees. The illustrated raised portion 160 creates a volume reduction of 4%.

With reference now to FIG. 13(c), the height (h) of the illustrated raised portion 160 is 6 mm. The length (w) of the raised portion 160 is 35 mm. The first angle (α) of the raised portion 160 is 15 degrees and the second angle (β) is 60 degrees. The illustrated raised portion 160 creates a volume reduction of 14%.

With reference now to FIG. 13(d), the height (h) of the illustrated raised portion 160 is 6 mm. The length (w) of the raised portion 160 is 18 mm. The first angle (α) of the raised portion 160 is 60 degrees and the second angle (β) is 60 degrees. The illustrated raised portion 160 creates a volume reduction of 8%.

FIGS. 14(a)-14(d) correspond to FIGS. 13(a)-13(d) and demonstrate the effect on the flow velocity in each configuration. FIGS. 14(a)-14(d) are flow velocity contours at a flow rate of 5 LPM. The contours of the no raised portion configuration of FIG. 14(a) clearly show that the elbow portion 162 creates an area of lower velocity gases, and it is in this area that the gases can cool. Each of the raised portions 160 of FIGS. 14(b)-14(d) has a positive effect on flow velocity, especially on velocities in the middle of the range. In addition, it can be seen that the more smoothly tapering asymmetric raised portions 160 of FIGS. 14(b) and 14(c) have significantly improved flow characteristics, because the shallow angle (α) of the tapering end of the raised portion 160 reduces the area of potential dead space downstream of the raised portion 160.

FIGS. 15(a)-15(d) correspond to FIGS. 13(a)-13(d) and show reverse flow vectors extending from the planes of several cross sections downstream of the elbow portion 162.

These vectors are overlaid onto the flow velocity contours of FIGS. 14(a)-14(d). The vector lengths represent the magnitudes of negative flow velocity components (i.e., the velocities of flow in the reverse direction to normal flow), which are indicative of recirculation that results in cooling gases. From these illustrations, it is apparent that the regions having the reverse flow components also have the lowest velocities, so it stands to reason that reducing recirculation will improve overall flow velocity and decrease cooling of the gases. Each of the raised portions 160 reduces recirculation relative to the configuration with no raised portion. The large asymmetric raised portion 160 of FIG. 15(c) appears to reduce recirculation the most. Again, the symmetric raised portion 160 of FIG. 15(d) does not appear to improve flow characteristics as much as the asymmetric raised portions 160 of FIGS. 15(b) and 15(c).

FIGS. 16(a)-16(c) correspond to FIGS. 13(a)-13(c); there is no corresponding illustration for FIG. 13(d). FIGS. 16(a)-16(c) show streamlines coloured by velocity magnitude and indicating paths taken by a zero mass particle if that particle were traveling with the flow. As shown in these illustrations, the recirculation and low velocity zones are greater in size following the sharp corner 170 of the elbow portion 162 where there is no raised portion. In addition, the larger raised portion 160 of FIG. 16(c) appears to improve performance over the smaller raised portion 160 of FIG. 16(b).

FIG. 17 is a plot illustrating the flow velocity components in a 5 LPM flow relative to the direction of flow at the same cross section for each of the configurations of FIGS. 13(a)-13(d). Negative velocities indicate flow velocity components in the reverse direction compared to normal flow (i.e., recirculation). Each of the configurations demonstrates very similar velocities in the upper region of the cross-section (i.e., the right side of the plot) but there are significant differences in the lower region of the cross-section (i.e., the left side of the plot). The configuration of FIG. 13(a), which has no raised portion, demonstrates large regions of stagnation (near 0) and recirculation (below 0) before reaching the upper region and middle region, where positive flow is taking place. As shown, the larger raised portion 160 of FIG. 13(c) demonstrates the most improvement to reducing or eliminating recirculation and/or stagnant flow.

FIG. 18 is a plot comparing pressure magnitudes taken along a horizontal line that extends along an axis of the configurations of FIGS. 13(a)-13(d). As shown in the plot of FIG. 18, the pressure drop, and therefore the resistance to flow, remains very similar for all four of the designs of FIGS. 13(a)-13(d). As such, the benefits of recirculation reduction can be obtained without significantly impacting resistance to flow.

In addition to improving flow characteristics, the raised portions 160 provide the advantage of thwarting the flow of condensate back into the chamber 134. In configurations desiring to take advantage of this characteristic, the symmetrical raised portion 160 of FIG. 13(d) provides enhanced performance over the asymmetrical raised portions 160 of FIGS. 13(b) and 13(c). In addition, configurations as shown in FIGS. 19(a)-19(d) use different geometries to allow gravity to reduce the likelihood of condensate flowing back to the humidification chamber 134. In each of the embodiments in FIGS. 19(a)-19(c), either the conduit 106 or the connector 152 has a lower region before attaching to the port 112 of the humidification chamber 134. These regions may not inhibit gas flow, but these regions may allow pooling of any condensate formed while reducing the likelihood of the condensate flowing back to the humidification chamber 134. FIG. 19(d) uses an inverted form of the previous embodi-

11

ments, where a high region is located proximally with a lower more distal region following, to reduce the likelihood of any condensate from flowing back to the humidification chamber 134.

These embodiments may provide solutions for condensate formation and pooling that may occur as a result of using a horizontal connection mechanism between a humidification chamber and a conduit. It is recognised that other mechanisms to reduce the likelihood of condensate flowing back to the humidification chamber may be possible and are not excluded from the scope of the disclosed apparatus and systems.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.

Reference to any prior art in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that that prior art forms part of the common general knowledge in the field of endeavour in any country in the world.

The disclosed apparatus and systems may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.

Where, in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.

Where a value is used with a term of approximation, that number is intended to include the range of roundable values unless otherwise apparent from the context of use.

It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the disclosed apparatus and systems and without diminishing its attendant advantages. For instance, various components may be repositioned as desired. It is therefore intended that such changes and modifications be included within the scope of the disclosed apparatus and systems. Moreover, not all of the features, aspects and advantages are necessarily required to practice the disclosed apparatus and systems. Accordingly, the scope of the present disclosure is intended to be defined only by the claims that follow.

What is claimed is:

1. A humidification chamber for a respiratory humidification system comprising:

an inlet port defining at least a first portion of a gases flow path;

an outlet port defining at least a second portion of the gases flow path; and

a body positioned in the outlet port, the body reducing an internal volume of the outlet port, a maximum height of the body is between 15% and 30% of a downstream diameter of the outlet port.

12

2. The humidification chamber of claim 1, wherein the body is asymmetrical along a length of the body, the length of the body extending in a direction of the gases flow path.

3. The humidification chamber of claim 1, wherein a maximum length of the body is greater than a maximum width of the body, the maximum length of the body extending in a direction of the gases flow path.

4. The humidification chamber of claim 2, wherein the body comprises a raised section extending smoothly towards a tapered downstream edge.

5. The humidification chamber of claim 1, wherein the outlet port comprises a first section extending in a first direction and a second section extending in a second direction, wherein the second section is downstream of the first section.

6. The humidification chamber of claim 5, wherein an interface between the first section and the second section comprises an abrupt change in direction of the gases flow path.

7. The humidification chamber of claim 5, wherein the first direction is approximately perpendicular to the second direction, such that the outlet port comprises an elbow portion.

8. The humidification chamber of claim 5, wherein the first section comprises an opening configured to receive a sensor.

9. The humidification chamber of claim 6, wherein an upstream edge of the body is positioned near the interface, the body forming a smooth transition between the first section to the second section for the gases flow path in use.

10. The humidification chamber of claim 1, wherein the body fills or at least partially fills a dead space region in the gases flow path in use.

11. The humidification chamber of claim 1, wherein the body reduces regions of low velocity along the gases flow path.

12. The humidification chamber of claim 1, wherein the body reduces recirculation of gases within the outlet port.

13. The humidification chamber of claim 1, wherein the body reduces stagnant flow within the gases flow path.

14. The humidification chamber of claim 1, wherein the body reduces an amount of condensate generated within the gases flow path in use, wherein the body reduces liquid flowing from a conduit connected to the outlet port to the humidification chamber in use.

15. The humidification chamber of claim 1, wherein the body reduces regions of low temperature along the gases flow path.

16. The humidification chamber of claim 1, wherein an angle between an upstream edge of the body and the outlet port is between 30 degrees and 60 degrees.

17. The humidification chamber of claim 1, wherein the body comprises an insert configured to be removably attached to the outlet port.

18. The humidification chamber of claim 1, wherein the body provides a volume reduction in the outlet port of approximately 14%.

19. The humidification chamber of claim 1, wherein the body is integrally formed with the outlet port.

* * * * *